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Spasticity

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SUMMARY PEARLS

- Spasticity chiefly refers to muscle tightness in response to stretch, although it has become a catch-all word to describe the clinical phenomenon of the upper motor neuron syndrome, including other impairments that lead to hypertonia and dysfunction.
- Spasticity results not only from central nervous pathology but also from concomitant and consequential intramuscular changes.
- Spasticity assessment should include the impact at rest, on well-being, and function.
- There are multiple pharmacologic and nonpharmacologic treatment options for spasticity, and a multimodal management approach is recommended for best clinical outcome.
- The resurgence of surgical treatment may reflect the recent recognition of personalized treatment rather than a stereotypical step approach.
- Emerging treatments for spasticity include new medications, cryoneurolysis, various physical modalities, and neuromodulation interventions.

Introduction

Spasticity is a common complication of many neurologic conditions. Estimates of spasticity incidence and prevalence vary because of the lack of a strict definition and clinical measurement of spasticity. It is estimated to occur in around 80% of persons with multiple sclerosis (MS) at some point of the disease,¹⁹⁵ and 65% to 78% in those with spinal cord injury (SCI).¹⁶⁷ There is little known about the occurrence in persons with traumatic brain injury (TBI), but in those with severe brainstem involvement, spasticity can exist in up to 40%.²⁷¹ Prevalence in stroke is about 20% to 40%.²⁷⁴ Within the first year of stroke, spasticity was found in 38% of survivors.²⁵⁴ However, spasticity is present in 97% of chronic stroke survivors with moderate to severe motor impairments.²¹³

Spasticity is significant because it predisposes to other complications, such as joint contractures and joint deformities, but as a comorbidity spasticity amplifies the effects of weakness and other motor disorders and contributes to limitations in activity and participation. Although it has become common practice to label any condition presenting as muscle tightness “spasticity,” it is important to point out that it is only one of the myriad consequences of the upper motor neuron (UMN) syndrome. Other important UMN syndrome features include weakness, exaggerated stretch reflexes, clonus, impaired coordination, and motor control/planning.¹⁶⁵ These numerous abnormalities evolve over time and interact with each other to produce a dynamic picture of varying clinical presentations after a UMN lesion.^{95,96} For example, weakness tends to result in immobilization in a shortened muscle length and predisposes to contracture. This in turn exacerbates the development of spasticity in the same muscles.

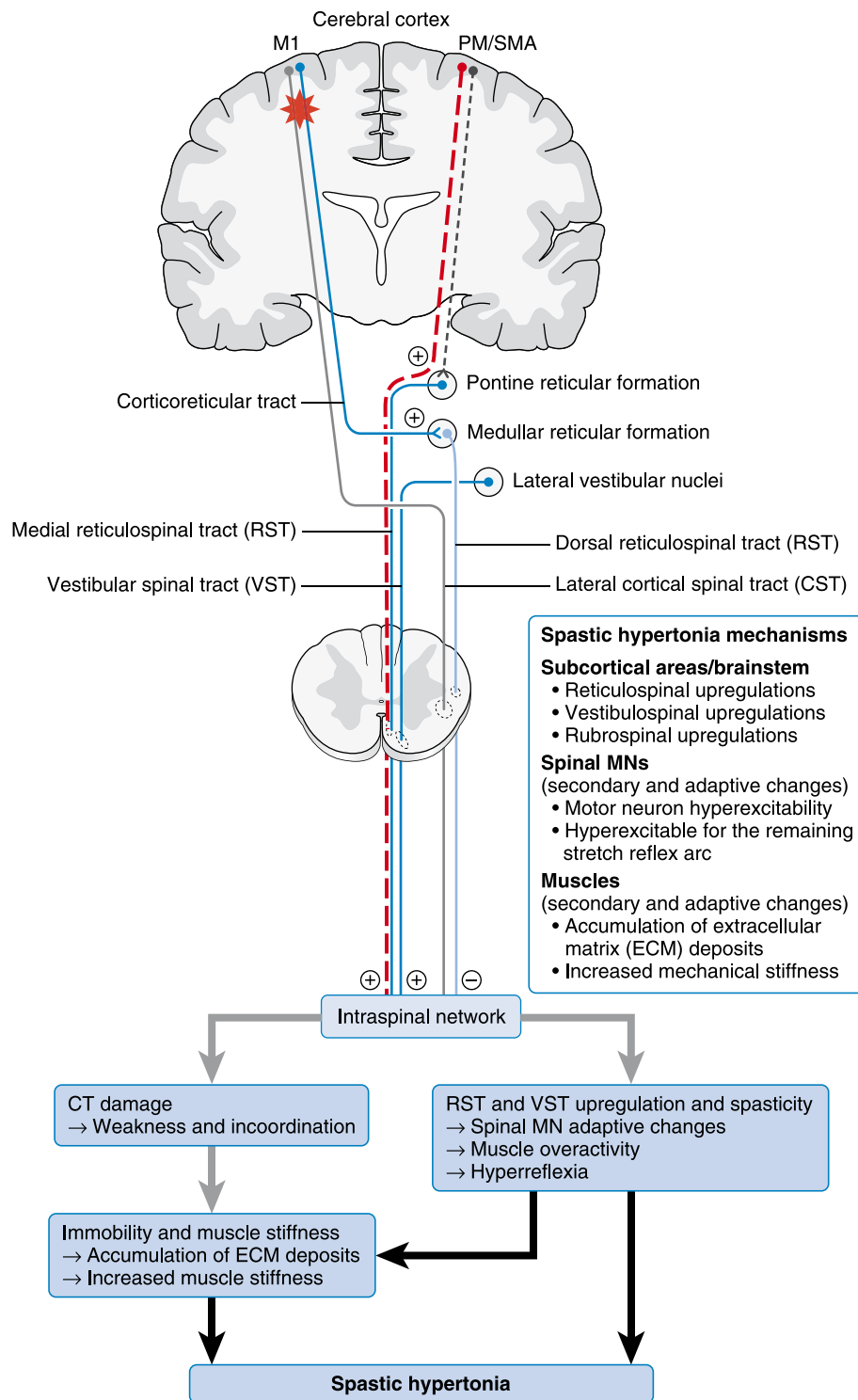
Further, spasticity exacerbates contracture and triggers a vicious cycle that worsens the condition.^{95,96,190}

Pathophysiology

The pathophysiology of spasticity is still not fully understood, but its understanding has evolved since Lance described the characteristics of spasticity in 1980.¹⁴⁵ This served as the basis of the most widely used definition of spasticity even though it is not validated. Since then, new definitions of spasticity have been proposed as research evidence and clinical understanding of this phenomenon progressed. Consequently, the Lance definition has been challenged and alternative definitions proposed.^{194,271} To reflect recent research advances in the underlying pathophysiology of poststroke spasticity and its relations with other parallel motor impairments, as outlined in the following sections, Li et al.¹⁵⁴ proposed a new definition of poststroke spasticity in 2021:

“Spasticity is manifested as velocity- and muscle length-dependent increase in resistance to externally imposed muscle stretch. It results from hyperexcitable descending excitatory brainstem pathways and from the resultant exaggerated stretch reflex responses. Other related motor impairments, including abnormal synergies, inappropriate muscle activation, and anomalous muscle co-activation, coexist with spasticity and share similar pathophysiological origins.”

Pathophysiologic changes and clinical presentations vary depending on the etiologies of damage. We first briefly summarize current understandings of pathophysiology for poststroke spasticity.^{33,38,96,151,153,184,187,215} As shown in Fig. 24.1, damages to motor cortex and its descending corticospinal tract (CST) occur as a result of a stroke. These damages cause muscle weakness (usually



• **Fig. 24.1** Schematic illustration of pathophysiology of spastic hypertonia. A stroke often leads to damages to the motor cortex and its descending corticospinal tract (CST). Muscle weakness and incoordination occur immediately after CST damages, subsequently leading to joint immobilization. Because of its anatomic proximity to CST, corticoreticular pathways to medullary reticular formation are damaged as well (red asterisk), resulting in diminished inputs from the dorsolateral reticulospinal (RST) tracts into the spinal motor network. Since the dorsolateral RST provides inhibitory inputs, its impairment makes the descending excitatory inputs (medial RST and VST) unopposed and becomes hyperexcitable overtime (red dotted line). RST hyperexcitability appears to be the most likely mechanism related to poststroke spasticity. RST hyperexcitability provides unopposed excitatory descending inputs to spinal stretch reflex circuits, resulting in elevated excitability of spinal motor neurons and spinal reflex circuits, thus spasticity and hyperreflexia. Clinical presentation of spasticity is confounded by concomitant changes in peripheral muscle properties, such as extracellular matrix (ECM) deposition and macromolecular crowding of hyaluronan and increased muscle stiffness.

unilateral) immediately after stroke, subsequently resulting in incoordination and often leading to joint immobilization. On the other hand, neuroplasticity occurs after stroke as well. Because of lesions of corticobulbar pathways accompanied with lesions of motor cortices and/or descending CST, bulbospinal hyperexcitability develops over time secondary to loss of cortical inhibition (i.e., disinhibition, or unmasking effects). Such hyperexcitability has been observed in several pathways, including medial reticulospinal, vestibulospinal, and rubrospinal projections.^{153,178,193} Medial reticulospinal hyperexcitability appears to be the most likely mechanism related to poststroke spasticity. It provides unopposed excitatory descending inputs to spinal stretch reflex circuits, resulting in elevated excitability of spinal motor neurons. This adaptive change can account for most clinical findings (e.g., exaggerated stretch reflex, velocity-dependent resistance to stretch, muscle overactivity, or spontaneous firings of motor units). As mentioned earlier, muscle overactivity also potentiates muscle shortening and weakness in a vicious cycle. Their interactions also facilitate limb immobilization, development of muscle and tendon contractures, and accumulation of extracellular matrix deposits.²¹⁵ Taken together, three different components have been proposed to account for increased resistance of spastic muscles or spasticity (see Fig. 24.1). These components are neither adequately distinguished in clinical examinations²⁵¹ nor in laboratory settings.¹⁶²

- *Hyperreflexia and reflex stiffness.* Clinically, hyperreflexia and reflex stiffness present as the commonly observed phenomenon of velocity-dependent increase in resistance to passive muscle stretch. It is mediated by hyperexcitability of stretch reflex circuits as a result of unopposed reticulospinal hyperexcitability.
- *Active muscle stiffness (hyaluronan accumulation).* Hyaluronan is the primary component in the extracellular matrix.⁸⁴ Its primary function in the muscle is to smooth the muscle fiber gliding system and to facilitate muscle force transmission (i.e., lubrication).⁸⁴ Immobilization or paresis decreases the normal turnover of the extracellular matrix, thus increasing its concentration within and between the muscular compartments, subsequently increasing the molecular weight and macromolecular crowding.^{54,188} These changes can increase the fluid viscosity,¹³⁷ thus decreasing the gliding and lubrication between the layers of collagen and muscle fibers. This change may be perceived by the stroke survivor as stiffness.⁴
- *Passive muscle stiffness (muscle fiber changes and connective tissue fibrosis).* Because of immobilization, spastic muscles may rest at a shortened length and muscle fibers become more stiff as a result.⁸⁵ Furthermore, fibrosis within the muscle tissue could be developed gradually secondary to the deposition of collagen within and between muscle bundles later on in the chronic phase.²⁴ These changes increase passive, mechanical stiffness of the muscle.

The previously mentioned components of spastic hypertonia are present after neurologic impairments of different origins. However, clinical characteristics of spasticity vary according to lesion locations. The pathophysiology of incomplete and complete SCI is also mediated by different mechanisms. In incomplete spinal cord lesions, both the CST and the dorsal reticulospinal tract (RST) are often damaged in the dorsolateral cord. Damage to the dorsal RST could also lead to a release phenomenon, producing flexor spasms and hyperreflexia, in addition to spasticity. In complete SCI, the situation is different. Both excitatory and inhibitory supraspinal descending inputs to spinal interneuronal network are absent, and thus both stretch reflex and flexor reflex afferents are completely disinhibited. As such, a strong flexor pattern follows, resulting in paraplegia with predominant flexion features.^{184,237}

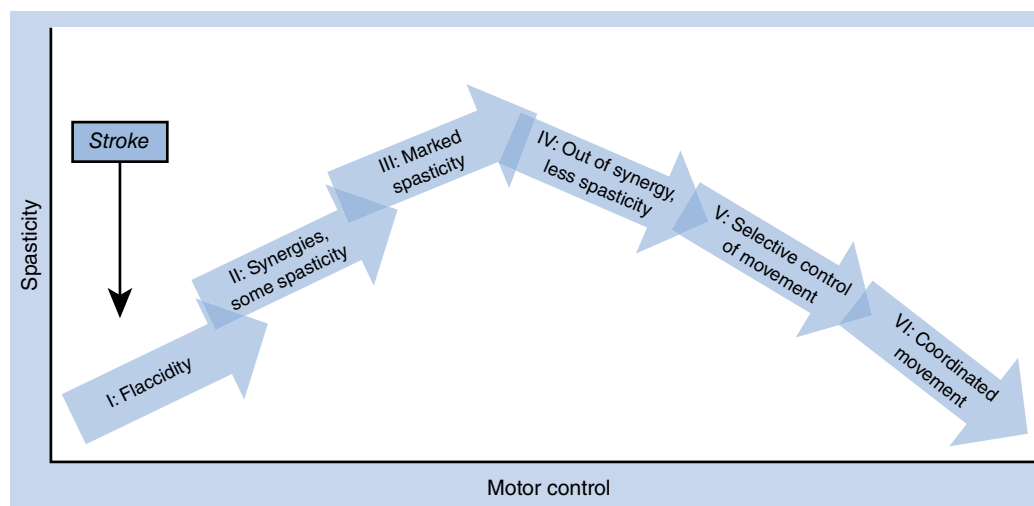
Motor recovery commences almost immediately after the onset of a central nervous lesion, when reversible changes resolve. For instance, after a stroke regardless of type (hemorrhagic or not) or location (cortical or subcortical), a relatively predictable pattern of recovery sets in.²⁵⁰ Brunnstrom^{34,35} empirically describes the stereotypical stages of motor recovery from flaccidity to full recovery of motor function (Fig. 24.2). As a stroke survivor recovers, progression from one recovery stage to the next toward recovery of normal movement occurs in an orderly manner, but evolution may be arrested at any stage.^{35,250}

Interestingly, as motor recovery progresses, spasticity decreases. The same pattern of motor recovery and emergence and disappearance of spasticity may not be seen in different etiologies of the UMN syndrome, however. What is common is the observation that there is a period of “shock” after the initial injury (traumatic or acquired), which is followed by a gradual return of reflexes, but not a sudden progression to hyperreflexia. This implies that there must be some sort of neuronal plastic changes after the initial injury. This process occurs at any time but is usually seen between 1 and 6 weeks after the initial injury.¹¹ Plastic rearrangement occurs within the brain and spinal cord^{11,38,95,96,184,187,189} and is regarded as an attempt at restoration of function through emergence of novel neuronal circuitry. This process of plastic rearrangement often results in muscle overactivity and hyperreflexia, and thus spasticity.⁷⁰ In a longitudinal study that examined the time course of development of spasticity and contractures at the wrist after 6 weeks of stroke, the authors reported that patients who recovered arm function showed signs of spasticity at all assessment points but did not develop contractures. In contrast, patients who did not recover useful arm function had signs of spasticity and changes associated with contracture formation tested over a follow-up course of 36 weeks¹⁶³ suggesting that spasticity, a harbinger of contractures, persisted. Collectively, emergence and disappearance of spasticity in the course of complete motor recovery implies that presence of spasticity reflects a phenomenon of abnormal plasticity. Spasticity may be maintained after its emergence if further plastic rearrangement and recovery are arrested. That spasticity is a manifestation of maladaptive plasticity is further supported by recent studies demonstrating that abnormal cortical reorganization can be modulated by interventions directed at decreasing spasticity.¹¹⁸

Clinical Presentation, Assessment, and Goal Setting

Problem Identification

When a patient presents with tight muscles and limb or joint deformities, and complains of an inability to perform certain tasks, it is often tempting to attribute all problems to spasticity. The clinician should keep in mind that spasticity is but one of the multitude of problems resulting from a UMN disease. Often weakness, rather than spasticity, is the primary cause of limitations. Patients rarely express their complaints as spasticity. Rather, they typically use words that describe the manifestations of spasticity, such as “muscle tightness,” “a limb that could not be moved,” or “spasms”²¹ or more often the resultant functional limitations, such as the inability to release a grasped object or difficulty with walking caused by an in-turned foot. Also, patients usually complain of symptoms resulting from other UMN abnormalities, such as spasms or clonus triggered by walking, weakness,



• **Fig. 24.2** Brunnstrom stages of motor recovery after stroke.

and difficulty manipulating objects caused by decreased hand dexterity. On the other hand, patients may not realize that they benefit from spasticity until it decreases or resolves. For example, extensor spasticity in the lower extremity facilitates transfer via stand-pivot (and patients may falsely equate this ability to preserved muscle strength). Similarly, involuntary synergistic movement of spastic muscles may bring psychological benefits to patients that they still experience movement in a paretic limb. Other potential benefits may include facilitation of circulation, orthostatic blood pressure support, and maintaining muscle mass. Thus obtaining a thorough, yet focused, history is of paramount importance in guiding the examination and formulation of mutually agreed upon treatment goals and plans.

Postural Abnormalities

Although the clinical presentation varies widely across individuals within and across patient populations, certain postural patterns are commonly observed. They are manifestations of imbalance of agonist and antagonist strength and hypertonia and their interactions with ground reaction force.¹⁶⁰ Thus a flexed elbow posture is not necessarily caused solely by flexor muscle group hypertonia but may be a combination of hypertonic flexors and weak extensors; or it could also be that both flexor and extensor muscle groups are both hypertonic, but the former predominates. It has been suggested that knowledge of the abnormal postural patterns helps inform goal setting, medication dosing, and outcome assessment of interventions, such as chemodenervation.¹¹⁰ Table 24.1 lists muscles that, when hypertonic, are potentially involved in commonly observed abnormal postures.

Impaired Movement

Just as it is with abnormal postures, impaired movements usually result from interactions between spasticity, weakness, and other features of the UMN syndrome, such as dystonia, loss of coordination and dexterity, or sustained contraction of muscles.

Functional Limitation

Activity limitations are even more complex because the causes of impaired movement are further aggravated by abnormalities other

than the UMN disorder but that are a direct result of the underlying disease. Tactile and proprioceptive sensory loss, visual field cut and hemineglect, and cognitive difficulties, such as learning a novel task and procedural sequencing, can magnify the motor challenges imposed upon by spasticity and weakness. Thus a rehabilitation effort that will be effective in enhancing activity participation should address not just spasticity and deficits in strength and coordination but concurrently tackle the related impairments.

CASE 24.1

Problem Identification

OA is a 50-year-old male who sustained a nonpenetrating brain injury from a motorcycle crash 16 years ago. He complains of difficulty reaching for objects because “it feels like something is holding my elbow back” and of foot drop that causes him to trip occasionally, especially after walking a long distance. He used to take oral baclofen 10 mg three times daily, which helped relax his arm and allowed easier reaching. However, he also thought that he has been tripping more often since starting baclofen and elected to discontinue the drug 3 months earlier. At rest, he held the right elbow in 30 degrees of flexion and the wrist and hand in normal anatomic position. An attempt to extend the right elbow with the shoulder abducted to around 70 degrees was characterized by coarse jerking (alternating elbow flexion and extension) and clenched fist. It took him about 30 seconds to extend the elbow to -20 degrees from the resting position. He had difficulty dorsiflexing the right ankle, being able to move it only through about a 20-degree range. On passive ranging, the right ankle reached only 20 degrees of dorsiflexion while the knee was in either extended or flexed position. Ashworth scale scores were 1 in the right shoulder adductor, elbow extensor, knee extensor, and ankle plantarflexor; 2 in the right elbow flexor; and 0 elsewhere. Based on these clinical findings alone, the clinician thought that the inability to reach was most likely caused by abnormal cocontraction of the elbow extensors (agonist) and flexors (antagonist) and the inability to grasp caused by synkinesis, or associated reaction, of the finger and thumb flexors triggered by elbow extension movement. The clinician also linked the history of increased tripping when the patient was receiving baclofen with the observation that the patient had right ankle dorsiflexor weakness and plantarflexor contracture. Thus in this patient, the two presenting problems, inability to reach and tripping, were most likely not caused by spasticity but other UMN disease features, such as weakness, abnormal agonist-antagonist cocontraction/loss of reciprocal inhibition, and synkinesis.

**TABLE
24.1****Common Postural Abnormalities Caused by Spasticity and Potential Muscle Involvement**

Postural Abnormality		Muscles Potentially Involved
Shoulder adduction		Pectoralis major Latissimus dorsi Coracobrachialis (especially when shoulder is forward flexed)
Shoulder internal rotation		Latissimus dorsi Subscapularis Teres major Pectoralis major
Elbow flexion		Brachialis Biceps Brachioradialis Pronator teres
Elbow extension		Triceps Anconeus
Forearm pronation		Pronator teres Pronator quadratus
Wrist extension		Extensor carpi radialis Extensor carpi ulnaris
Wrist flexion		Flexor carpi radialis Flexor carpi ulnaris Palmaris longus
Metacarpophalangeal (knuckle) flexion		Lumbricals
Finger flexion		Flexor digitorum superficialis (proximal phalanx) Flexor digitorum profundus (distal phalanx)
Thumb flexion		Flexor pollicis brevis (proximal) Flexor pollicis longus (distal phalanx)

Continued

TABLE 24.1 Common Postural Abnormalities Caused by Spasticity and Potential Muscle Involvement—cont’d

Postural Abnormality		Muscles Potentially Involved
Trunk flexion, lateral		Quadratus lumborum Latissimus dorsi
Hip flexion		Psoas Iliacus Rectus femoris
Hip extension		Gluteus maximus
Hip adduction		Adductor muscle complex
Knee extension		Quadriceps complex
Knee flexion		Hamstrings Gastrocnemius

**TABLE
24.1****Common Postural Abnormalities Caused by Spasticity and Potential Muscle Involvement—cont'd**

Postural Abnormality		Muscles Potentially Involved
Ankle plantarflexion		Gastrocnemius Soleus Tibialis posterior Flexor digitorum longus Flexor hallucis longus
Ankle inversion		Tibialis posterior Tibialis anterior Extensor hallucis longus
Small toe flexion		Flexor digitorum brevis (proximal) Flexor digitorum longus (distal)
Great toe hyperextension		Extensor hallucis longus Ankle plantarflexors

Clinical Assessment

Spasticity assessment typically consists of a combination of quantitative and qualitative measures. Although it is true that quantitative measures are desirable because of their inherent objectivity and reliability, they may not be practical and discourage clinicians from assessing and managing spasticity. Ideal measures include biomechanical and electrophysiologic tests, but many of the devices needed to carry these out are not available to a typical clinician, and time needed to perform them properly may impose excessive demands in a busy practice. Thus a combination of clinical measures, some of which appear to correlate with biomechanical and electrophysiologic assessment, is a practical approach. Table 24.2 summarizes many of these evaluation tools and techniques.

Platz et al. proposed a rating scale coined REsistance to PASsive movement, or REPAS.²⁰⁷ Test items for various passive limb motions based on the Ashworth scale were analyzed and compared with other scales that measure spasticity and related phenomena, motor impairment, and activity limitation. This summary rating scale was found to be highly internally consistent and reliable for the clinical assessment of resistance to passive movement in patients with UMN disorders. Further, the REPAS scores were associated with a patient's functional abilities and a caregiver's difficulty in assisting with hygiene. The REPAS scale offers an alternative that may be able to more accurately assess spasticity and effects of therapeutic interventions clinically. In

Box 24.1 and Table 24.3, we propose a systematic approach to history taking and clinical assessment of spasticity that can be modified to suit different clinical scenarios. REPAS, or more popularly used scales, such as the Ashworth, can be made part of this proposed examination sequence.

Goal Setting

As described earlier, patients may benefit from spasticity treatment even without knowing it. An important component of assessment and management decision-making is to determine whether the patient and caregivers can benefit from the treatment (i.e., any goals of potential treatment). It is also important to arrive at treatment goals that are mutually agreed upon by the patient (or caregiver) and clinician. Identifying goals a priori provides a context for identifying pertinent problems and their solutions while also managing utilization of resources. It is not uncommon for patients to desire goals of regaining normal form and function, but because this is not always achievable, a discussion regarding goal setting before initiating treatment can help manage expectations of treatment outcomes. Goal setting can also help identify the best treatment strategy at a particular time in the course of a person's rehabilitation and recovery. A useful method uses the SMART acronym originally used in the business sector.⁶³ Each of the letters in the acronym stands for a criterion that characterizes goals and objectives and that has appeared in various

TABLE 24.2 Measures of Spasticity Categorized Based on WHO International Classification of Functioning, Disability and Health

Domain	Measure	Examples
Body function and structure	Clinical	Visual analog scale Goniometer Ashworth scale (and its modified version) ^b Tardieu method Tone assessment scale Spasm severity scale Spasm frequency scale
	Physiologic	Hmax/Mmax ratio Vibratory inhibitory index Tendon reflex gain Angular joint velocity Reflex threshold angle Torque measurement Pendulum test Myotonometer
Activity	Impairment ^a	Fugl-Meyer assessment Jebsen-Taylor hand test Wolf motor test Action research arm test Berg balance scale
	Function ^a	Functional independence measure Barthel index Disability assessment scale ^b Frenchay arm test Timed walking test Motion analysis Individualized functional tasks
Participation	Quality of life ^a	SF-36 Health Survey Satisfaction with life scale EuroQol (EQ-5D)

^aMeasure is not specific for spasticity.
^bMeasure was developed specifically for spasticity.
WHO, World Health Organization.

CASE 24.2

Clinical Assessment and Goal Setting

WT is a 49-year-old right-handed female administrative assistant who had a left basal ganglia hemorrhage caused by hypertension 1 year earlier. She presented with tightness of her right index finger flexors. Going deeper into the nature of her presenting concern, the clinician learned that although the patient is able to actively flex her fingers, when she attempts to type, she is unable to extend the right index finger to lift the digit off the keyboard. The clinician also inquired about how else the tightness of her finger has affected her ability to do things. To this, the patient expressed her concern that her limited typing ability may be grounds for dismissal from work because preparing written communication is an essential function of her current job. Upon further questioning, the clinician also found out that the patient has difficulty extending her right elbow each time she attempts to reach, and that she has no other problems related to muscle tightness. The clinician's examination consisted of observing the patient actively flexing and extending the right elbow. He noticed that the patient slowly but smoothly flexed her right elbow, but that she struggled as she attempted to extend the same elbow, often jerking the arm. The patient was not able to actively extend the right elbow against minimal resistance. The patient demonstrated her ability to actively flex the right digits, but difficulty extending the index and middle fingers. Ashworth scale testing of the right upper limb yielded the following results: shoulder adductors and internal rotators: 1, elbow flexors: 1, elbow extensors: 0, wrist flexors and extensors: 0, finger and thumb flexors: 0.

The clinician suggested injecting botulinum toxin injection to the right elbow flexors and the right index and middle finger flexors. The patient asked the clinician why the shoulder adductors will not be treated because she heard that the "score was a 1, while the fingers had a score of 0." The clinician confirmed with the patient that she has no trouble cleaning her underarm or donning shirtsleeves, and he thus felt that muscle tightness alone is not a reason enough for treatment. Intervention should be considered only if spasticity causes or predisposes to the development of problems and complications. Furthermore, the clinician pointed out that the Ashworth score, which is done passively, is not as important as the patient's report of functional problems with the two finger flexors that tend to tighten up in relation to a particular task. The patient also asked about other treatment options. The clinician replied that he did not think that a problem that is limited to two muscle groups in a single limb justify the systemic side effects of oral drugs. He also pointed out that another injection technique using phenol carries more risks for adverse events than botulinum toxin. The clinician and patient mutually agreed on the following goals, treatment plan, and measure of success:

CASE 24.2

Clinical Assessment and Goal Setting—cont'd

Problem	Goal	Treatment Approaches	Outcome Measure	Target Date
Difficulty reaching caused by spastic elbow flexors and, possibly, weak elbow extensors	Increase active elbow extension to at least -30 degrees to allow positioning of forearm close to keyboard	Botulinum toxin chemodenervation to brachialis, biceps, and brachioradialis, immediately followed by strengthening of the elbow extensors and functional reaching exercises	Technical: Elbow flexor Ashworth score 0; increased smoothness of elbow extension-flexion Functional: Ability to actively extend elbow far enough to reach the keyboard	2 months postinjection with botulinum toxin
Unable to type with index finger caused by tight flexor	Type with right index finger	Botulinum toxin chemodenervation to flexor digitorum superficialis and profundus, immediately followed by task-specific home program consisting of practicing typing with index finger	Technical: Finger flexor Ashworth score 0; recovery of isolated active index finger flexion and extension Functional: Ability to type using index finger (short term) with 100% accuracy in hitting desired keyboard tiles (long term)	2 months postchemodenervation (short term); may need to have a re-injection and further therapeutic exercises to achieve long-term outcome (4–5 months after first chemodenervation)
Minimally tight shoulder adductors and internal rotators	Not applicable (n/a)	No intervention, as the tightness of the shoulder is not causing any difficulty currently	n/a	n/a

• BOX 24.1 Important Historical Points in Spasticity Assessment

Is the limb tight all the time or only at certain times?
 Does a particular position or movement trigger tightness?
 Is the tightness related to spasms?
 Does the tightness cause pain?
 Have there been episodes of skin compromise due to tightness or spasm?
 Does the tightness result in difficulty with cleaning?
 Does the tightness result in difficulty donning splints?
 Does the tightness limit ability to move limbs, reach for objects, and use the hands?
 Does the tightness of the lower limbs result in problems with transferring from one surface to another or with walking?

What treatments for muscle tightness have been tried previously and their outcome?
 What are the current medications?
 Was there a recent increase in tightness (that may warrant further diagnostic testing to rule out a new neurologic problem)?
 Any recent medical problems?

From Francisco GE, Li S. Clinical assessment and management of spasticity and contractures in traumatic brain injury. In Pandyan AD, Hermens H, Conway B, eds. *Spasticity and Contractures in Neurological Rehabilitation: A Guide to Clinical Practice and Research*. CRC Press; 2017.

TABLE 24.3 Practical Clinical Examination Sequence

Examination Phase	What to Look for	What Can Be Gleaned
Observation	Observe limb posture at rest and how they change with positions	Posture at rest—sustained muscle contraction (dystonia) or contractures Position-dependent postural changes—synkinesis, or associated reactions
Active (patient performs actions)	How limbs move and how much active range is available Pain and discomfort as the patient moves	Functional strength, coordination, spastic cocontraction, contractures, presence of other movement disorders
Passive (examiner performs maneuvers)	Passive range of motion, strength, muscle tone, velocity-dependent “angle of catch,” clonus Pain and discomfort as a maneuver is performed	Spasticity Rigidity Contracture
Functional	Gait characteristics and associated upper limb and trunk postural abnormalities Performance of specific tests and tasks (both formal tests, such as Frenchay and improvised tasks such as demonstrating ability to pick up a bottle of water and pour its contents to a cup)	Impact of multiple impairments (e.g., spasticity, weakness) on performance

iterations to fit different purposes. For the purpose of goal setting in spasticity management, SMART—or SMARTER—can stand for:

- S: Specific (well defined and targets a specific problem to be addressed)
- M: Measurable (either quantitatively, as for technical goals, or qualitatively as for symptom-directed goals); Meaningful (achievement of goal should be beneficial to the patient or caregiver)
- A: Agreed upon (the patient or caregiver and clinician work toward a common end)
- R: Realistic (will the patient's potential for improvement and available resources support achievement of treatment goal?)
- T: Time bound (achievement of goal should be within a reasonable amount of time)
- E: Evaluated (at predetermined points in time, goal achievement and progress in doing so should be performed to determine effectiveness of intervention)
- R: Revised (based on evaluation of goal achievement, new treatment goals may be identified, or prior ones revised)

Management

Nonpharmacologic

Although spasticity is a neurologically based condition, its obvious manifestation is physical, and thus physical modalities are a mainstay in the first line of treatment because they are widely available and innocuous relative to drugs. Passive stretching has been shown to be effective in reducing tone and increasing range of motion (ROM) in patients with spasticity after brain injury.²⁴³ Splinting and casting are often used in the acute setting for sustained stretching.^{23,28,182,209,212} Casting alone seems sufficient to prevent contracture and reduce spasticity if the intervention is initiated early after severe brain injury. A systematic review on the use of upper extremity casting found high variability in casting protocols, which indicates no consensus in technique.¹⁴⁶ Individualized stretching has been shown to be clinically promising to reduce wrist and finger spasticity and improve passive ROM.⁵¹ More recent studies have shown some effects on spasticity reduction after stroke.^{142,252} The efficacy of electrical stimulation on spasticity reduction is not well established.⁴⁴ A recent review³⁶ suggested that moderate-quality evidence exists for transcutaneous electrical nerve stimulation, neuromuscular electrical stimulation, resistance training, and lower extremity ergometer training with or without functional electrical stimulation in decreasing poststroke spasticity. A novel technique demonstrated a long-lasting effect on spasticity reduction if electrical stimulation was triggered by voluntary breathing.¹⁵⁶ Guided self-rehabilitation contracts, a diary-based rehabilitation strategy, wherein specific muscles are subjected to “high-load, home self-stretching techniques,” is a promising recent addition to nonpharmacologic treatment strategies.^{98,99}

Pharmacologic

There is more than one way to approach the pharmacologic management of spasticity. Historically, a stepladder paradigm was useful, beginning with the least invasive treatment and culminating with surgical procedures should the nonoperative options fail. More recently, this sequential approach has been abandoned in favor of concurrent use of both noninvasive and invasive procedures (e.g., injection therapy concurrent with therapeutic

exercises, or using injection therapy for residual focal spasticity in persons whose generalized spasticity is simultaneously managed with intrathecal drugs). This reflects better appreciation of the magnitude of the problem—that is, spasticity alone does not account for presenting problem, and other features of UMN syndrome contribute significantly. Consequently, different treatments may be required to increase the chances of successful outcome. There are also different methods when deciding on pharmacologic intervention for spasticity. One approach is to base drug choice on the clinical presentation and the number of limbs involved. The algorithm in Fig. 24.3 illustrates this.

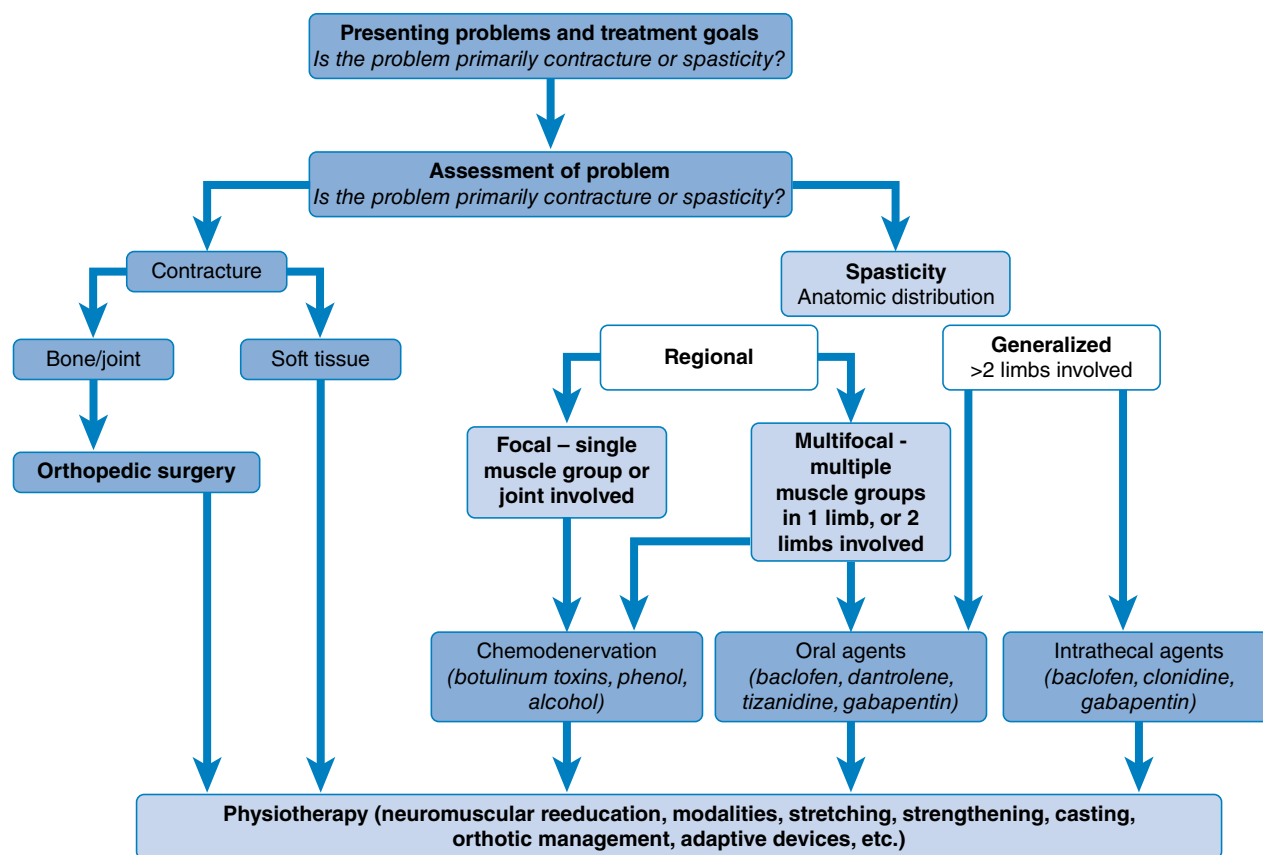
Another method is to determine the choice of drug on the underlying pathophysiology of spasticity, which in turn is influenced by neuroanatomic involvement in the disease process^{33,60} (Fig. 24.4). This can be particularly challenging because it is not always easy to determine the exact pathophysiologic process responsible for spasticity.

Oral Spasmolytics

Various medications with different mechanisms of action have been used to treat spasticity. The most commonly used are baclofen, a gamma-aminobutyric acid B (GABA-B)⁵⁷ agonist, tizanidine, an α_2 -adrenergic receptor agonist, and dantrolene, a hydantoin derivative (Table 24.4). Others include benzodiazepines, which are GABA-A agonists, and gabapentin, a likely selective inhibitor of voltage-gated calcium channels. Both are antiepileptic drugs (AED). Based on suggestions made by a few case reports or series, AEDs, including levetiracetam, pregabalin, phenytoin, and carbamazepine, appear to exert spasmolytic effects.^{29,109} Although their mechanisms of action vary, in general oral spasmolytics are associated with similar and related adverse events, such as sedation, drowsiness, and weakness. These side effects limit the effective utilization of oral spasmolytics in general. However, when dosed appropriately, oral spasmolytics can adequately control spasms and pain and facilitate such things as transfers and bed mobility in persons with SCI. A retrospective analysis that used medical and pharmacy claims data from a large, national US health plan speculated that poor medication adherence could be a function of the adverse effects associated with oral baclofen, tizanidine, or dantrolene, or of the perception of the limited efficacy of these drugs. Oral drug adherence did not vary across patient populations, including stroke, SCI, TBI, cerebral palsy (CP), and MS.¹⁰⁵

In general, the evidence for therapeutic efficacy for all spasmolytic agents is supported by only a few placebo-controlled trials. Consequently, there is no evidence-based treatment algorithm for drug choice and dosing. Instead, medication choice is based on practical reasons, such as favoring or avoiding certain medications, or in some situations capitalizing on certain known adverse effects. For instance, a person with SCI and painful nocturnal spasms may be best managed by tizanidine, which has analgesic properties independent of its spasmolytic effect. Because one of its common side effects is sedation, tizanidine may also help promote sleep that is disturbed by the spasms. However, if the same patient has orthostatic hypotension or is taking clonidine, another α agonist, tizanidine, may not be the best choice because of its potential hypotensive effect.

Medication dosages vary among persons largely because of the tolerability of side effects. Deficits in alertness and attention in persons with cerebral disease are magnified by the sedating effects of oral spasmolytics. It is not uncommon to bypass oral drugs in this patient population and instead use focal injection therapy as first-line treatment. Owing to the therapeutic effect of spasmolytics,



• Fig. 24.3 Clinical treatment decision-making algorithm for spasticity.

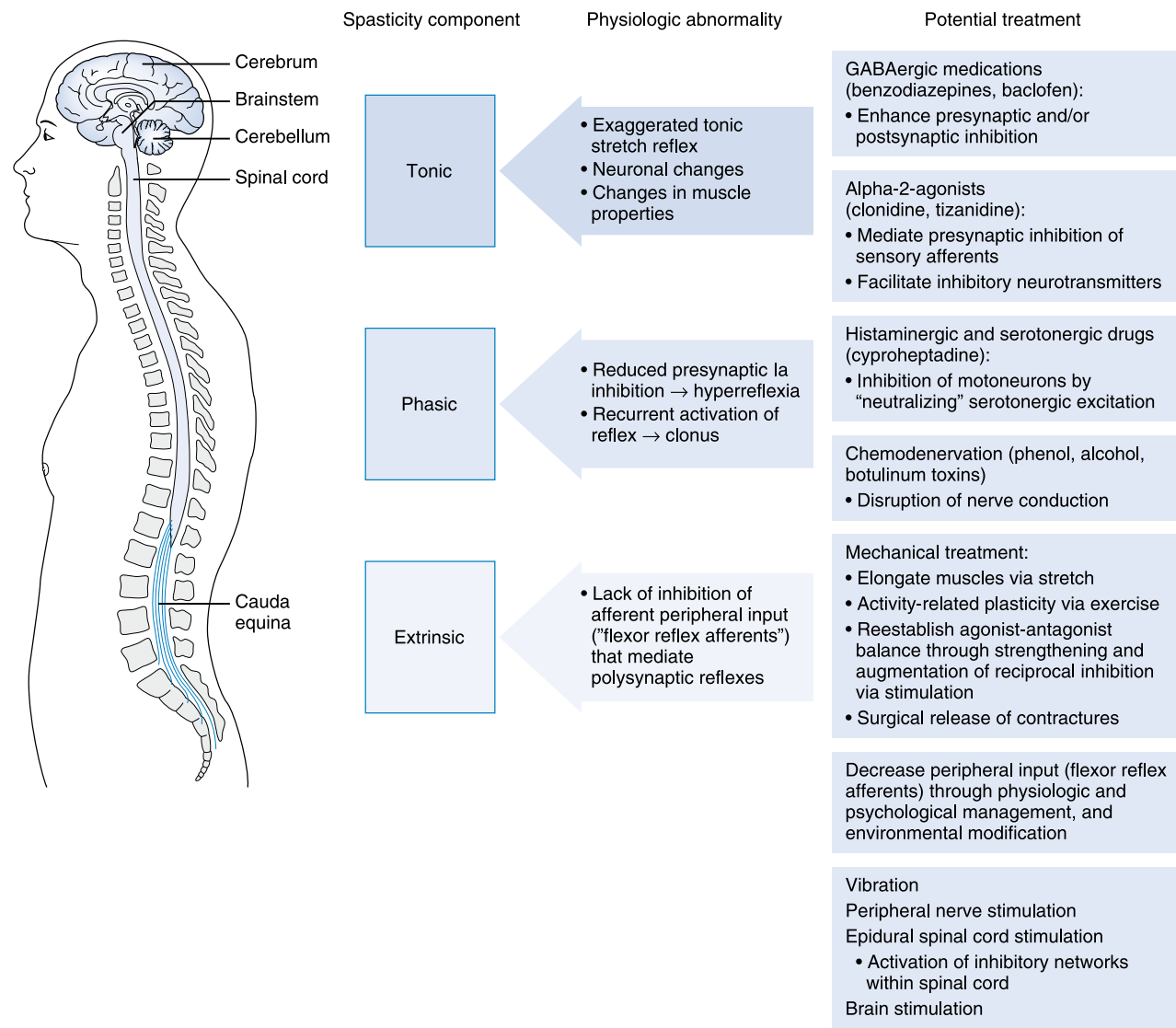
a reduction in muscle hypertonia may unmask latent weakness²⁴⁴ in persons with incomplete SCI or MS, who may lose the ability to ambulate should the weakness become pronounced. In this situation, a lower dose of the spasmolytic drug is preferred even though spasticity will not be controlled as well. Another option is to use a combination of different medications albeit at lower doses to circumvent a particular side effect typical of a drug.

As presented earlier in this chapter, there are various independent or interacting pathophysiologic processes that are postulated to cause spasticity. Yet, in the absence of solid scientific evidence, medication choices are typically empiric or to a certain extent based on a drug's side effect profile. This common practice may explain why a particular medication is not consistently effective both within and across patient populations. Corollary to this, the fact that different pathomechanisms are involved because of varying disease types and severity or anatomic involvement may explain why a single medication cannot be always effective for a given disease condition. This also imposes an important challenge in designing comparative efficacy and safety caused by the heterogeneity of the likely pathomechanisms underlying the spastic condition and the mechanisms of action of various spasmolytic drugs.

Baclofen is an analog of GABA, the most potent inhibitory neurotransmitter. It binds to GABA-B receptors that are widespread in Ia sensory afferent neurons and alpha motor neurons. It is the most widely studied oral spasmolytic in various patient populations, including SCI, MS, CP, and acquired brain injuries (TBI, stroke, anoxia, or encephalopathy).^{16,64,168,171,177,206} As with many spasticity medication trials, the studies involving baclofen demonstrated reduction in hypertonia and spasms but did

not investigate functional impact. Adverse effects of baclofen include drowsiness and weakness, which are shared by other oral spasmolytics. Abrupt discontinuation of baclofen may result in a withdrawal syndrome characterized by rebound spasticity, hallucinations, and seizures.

Tizanidine is believed to exert its effects by inhibiting the facilitatory ceruleospinal tracts and the release of excitatory amino acid from spinal interneurons. As a result, presynaptic interneuronal excitation is suppressed, manifesting as reduction in tonic and phasic stretch reflexes and agonist-antagonist cocontraction.²⁴⁴ Similar to baclofen, its efficacy in reducing muscle hypertonia and clonus, but not in improving function, has been demonstrated by various investigations.¹⁶⁹ However, in patients with stroke or TBI, tizanidine was deemed inferior to botulinum toxin in efficacy and safety in treating upper limb spasticity.²⁴¹ In addition to the typical side effects of oral spasmolytics, hepatotoxicity may also occur. Thus monitoring the liver function test is important, especially in those patients who concomitantly take hepatically cleared drugs. Being a central α_2 -adrenergic receptor agonist, tizanidine should be used carefully in patients with hypotension or are concomitantly taking other alpha agonists, such as clonidine. Likewise, caution should be used when coadministering tizanidine with fluoroquinolone antibiotics, which may increase the former's serum concentration. Tizanidine is a peculiar spasmolytic in that it has a dose-dependent antinociceptive effect presumably caused by a reduction in the release of substance P and activity of excitatory amino acids at the spinal level.²²² This makes tizanidine a suitable choice for those with concurrent spasticity and pain. Concerns for its side effects, mainly drowsiness, sedation, reduced



• Fig. 24.4 Spasticity treatment based on pathophysiology.

TABLE 24.4 Classification of Drugs With Spasmolytic Activity Based on Mechanism of Action

General Mechanism of Action	Drugs With Spasmolytic Activity
GABAergic	GABA _A (benzodiazepines, carisoprodol) GABA _B (baclofen)
Monoaminergic	Tizanidine Clonidine Cyproheptadine Cyclobenzaprine
Ion flux blockade	Dantrolene (blocks calcium) 4-aminopyridine (blocks potassium) Antiepileptic drugs (e.g., lamotrigine, phenytoin [blocks sodium]; pregabalin and gabapentin [blocks calcium])
Excitatory amino acid suppression	Orphenadrine
Others	Cannabinoids

attention, and memory impairment, and the potential for physiologic dependence, have limited the use of benzodiazepines as first-line treatment for spasticity. They appear to be used more often when spasticity is accompanied by other conditions that are also amenable to benzodiazepine therapy, such as seizures, anxiety, insomnia, spasms, and other movement disorders. Like baclofen, benzodiazepines exert their effects through modulation of GABAergic transmission, but unlike baclofen, which bind GABA-B receptors, benzodiazepines bind GABA-A receptors. Also similar to baclofen, abrupt discontinuation of benzodiazepines may result in a withdrawal syndrome.

Another GABAergic agent, gabapentin, is commonly used for the treatment of seizures and neuropathic pain, but through its selective inhibitory effect on voltage-gated calcium channels containing the $\alpha_2\delta_1$ subunit it decreases muscle hypertonia. A few studies have demonstrated gabapentin's beneficial effect on spasticity, although it seems that doses as high as 2400 mg/day are needed.^{57,65,78,102,183} Similar to other oral spasmolytics, the main adverse effects of gabapentin include drowsiness, sedation, and dizziness.

Dantrolene is used to manage a myriad of conditions, such as neuroleptic malignant syndrome and 3,4-methylenedioxymethamphetamine (Ecstasy) overdose, but its first reported therapeutic use was for spasticity.¹³³ Unlike baclofen and tizanidine, which act on the central nervous system, dantrolene works directly on skeletal muscle. Although it is widely regarded to exert its muscle relaxant effect by inhibiting the release of calcium from the sarcoplasmic reticulum during excitation-contraction coupling, its exact mechanism of action had not been elucidated until recently. It appears that a direct or indirect inhibition of ryanodine receptor, the major calcium release channel of the skeletal muscle sarcoplasmic reticulum, is fundamental in the molecular action of dantrolene in decreasing intracellular calcium concentration.¹³⁹ Several investigations have supported dantrolene's beneficial effect on spasticity.^{16,130,133,257} Although it is peripherally acting, dantrolene has also been associated with side effects that appear to be centrally mediated, such as drowsiness, dizziness, fatigue, and weakness, perhaps through alteration of neuronal calcium homeostasis.^{75,130} Because of its potential for hepatotoxicity, regular monitoring of liver function is recommended.

A systematic review of 101 randomized trials⁴⁶ found that baclofen, tizanidine, and dantrolene are effective in reducing spasticity relative to placebo, particularly in the MS population. It also found evidence that baclofen and tizanidine are equivalent in effectiveness, but there was insufficient evidence to compare with dantrolene. The review also found fair evidence that the overall side effect rate between baclofen and tizanidine is similar and that the former is more associated with weakness and the latter with dry mouth.

Other agents that have been observed to exert a spasmolytic effect in various patient populations with UMN diseases include the alpha-adrenergic agonist clonidine, which has more affinity to α_2 than α_1 receptors¹⁶⁰; 4-aminopyridine, a potassium channel blocker that is thought to enhance neural transmission by improving axonal conduction and synaptic neurotransmitter release, has also been shown to reduce spasticity in patients with SCI.²¹¹ Cyproheptadine, a serotonergic antagonist, has been demonstrated to decrease spasticity in persons with SCI.²⁵³ Cannabis and its active ingredient, delta-9-tetrahydrocannabinol (THC), and cannabinoids have been shown to decrease spasticity,^{52,218} especially in those with pain.¹⁰⁶ The best evidence of efficacy for spasticity is in the use of a THC and cannabidiol oromucosal spray (Sativex) in MS.¹⁹⁷ Its use is limited owing to concerns about potential cognitive impairment and possible increased risk of psychosis associated with cannabis.

Focal Treatment—Botulinum Toxins

Chemodenervation using botulinum toxins has become a widely used spasticity treatment. It is recommended for the management of focal spasticity²⁴² or when the treatment plan targets a particular muscle or group of muscles. Botulinum toxin exerts its effect through inhibition of acetylcholine release at the neuromuscular junction. The clinical effects of the toxin do not manifest until several days after an injection, largely because of a complex process involved, which includes (1) internalization of the toxin, (2) reduction and translocation of disulfide bonds holding the toxin's light and heavy polypeptide chains, and (3) inhibition of acetylcholine release. The last one occurs because the internalized toxin cleaves so-called SNARE (soluble n-ethylmaleimide sensitive fusion attachment receptor) proteins that facilitate neuronal membrane adherence of synaptic vesicles, a necessary step before extrusion into the neuromuscular junction. The clinical effects last about 3 months, which was previously thought to be caused by neuronal

resprouting. However, it has been shown that functional repair of the previously affected neuromuscular junctions is the likely mechanism for the duration of action.⁵⁹ Until recently, there had been only seven serotypes (A, B, C1, D, E, F, G) of *Clostridium botulinum* identified. A newer serotype, botulinum toxin X, was then identified and characterized,²⁷³ and the following year a botulinum neurotoxin-like toxin discovered in a commensal strain of *Enterococcus faecium* was reported.²⁷² At the current time only serotypes A and B are utilized clinically and have US Food and Drug Administration (FDA)-approved indication for spasticity: abobotulinumtoxinA, incobotulinumtoxinA, onabotulinumtoxinA, and rimabotulinumtoxinB. Recently, daxibotulinumtoxinA was approved by the FDA for cervical dystonia but not yet for spasticity. There are newer toxins, such as prabotulinumtoxinA, letibotulinumtoxinA, Meditoxin, Neuronox, and A2NTX,^{47,86,92,225,245} which will not be discussed here because they are primarily used for cosmetic indications or are not widely in use. All serotypes (except for the newly described toxin in an *Enterococcus* species) share common properties in that they are all clostridial neurotoxins with a bichain structure. They all inhibit acetylcholine release, and the muscle paralysis they produce is reversible. However, they are antigenically distinct, and they have varying neuronal binding sites and enzymatic actions on different SNARE proteins. They may also have different specificity. Nevertheless, the effects of all botulinum toxin A (BoNT-A) products are mediated by the 150 kDa BoNT-A neurotoxin. It was found that, at current FDA-approved doses, abobotulinumtoxinA contains greater amounts of active neurotoxin as compared with other BoNT-A products.⁷² Therefore it is important to highlight that different botulinum toxin products are not interchangeable and cannot be converted into units of other products.⁷¹ Table 24.5 summarizes these differences and other properties of pharmaceutical preparations of these toxins.

Consensus and review papers support the use of botulinum toxins for the management of spastic conditions in pediatric and adult populations.^{58,220,238,240,254} Although overwhelming evidence demonstrates that botulinum toxin chemodenervation results in significant improvement at the body function and structure level,^{10,12,19,39,144,220,221,235,238,240,246,265} it has not been shown unequivocally to be significantly efficacious in improving the activity and participation domains of the International Classification of Functioning, Disability and Health.^{6,31,41,76} Perhaps published studies failed to show impact on function caused by methodologic limitations, such as in subject selection and sensitivity of outcome measures.

Clinical issues related to botulinum toxin chemodenervation

Dosing. Clinical experience, regulatory and insurance coverage restrictions, and manufacturers' recommendations based on a few studies largely dictate choice of doses of the various botulinum toxins. There are a handful of dose-ranging studies that define dose-related therapeutic and adverse effects in spasticity.^{9,20,45,97,119,239} Dosages that are used in current practice, recommended by consensus statements²⁶⁵ and described in retrospective and observational studies,^{62,67,82} appear to be higher than doses used in published randomized controlled studies. Other than these guidelines and recommendations for dosing based on severity of spasticity and muscle size, there is no objective method of predicting response to a particular dose of botulinum toxin. In the future, techniques such as electromyography, ultrasound elastography, and quantitative modeling may assist in the assessment of spasticity's impact on muscle architecture and function and (eventually) with treatment decision making.^{42,88,201,204}

TABLE 24.5 Comparison of Commercially Available Botulinum Toxin Preparations

	AbobotulinumtoxinA	DaxibotulinumtoxinA	IncobotulinumtoxinA	OnabotulinumtoxinA	RimabotulinumtoxinB
Clostridium botulinum strain	Ipsen strain	Hall A	Hall A	Hall A	Bean B
SNARE target	SNAP-25	SNAP25	SNAP-25	SNAP-25	VAMP
Preparation	Powder for reconstitution	Powder for reconstitution	Powder for reconstitution	Powder for reconstitution	Ready-to-use solution
Storage	<8°C	20–25°C	<25°C	<8°C	<8°C
Shelf-life	15 months	36 months	36 months	24–36 months	24 months
pH of solution	7.4	5.5–5.6	7.4	7.4	5.6
Stabilization	Freeze-dried	Proprietary stabilizing excipient (peptide (RTP004) instead of human serum albumin	Vacuum-dried	Vacuum-dried	pH-reduced
Excipients	Human serum albumin 125 µg/vial; Lactose 2500 µg/vial	PS20, sugar, buffer, excipient peptide (RTP004)	Human serum albumin 1 mg/vial; sucrose 5 mg/vial	Human serum albumin 500 µg/vial; sodium chloride 900 µg/vial	0.05% human serum albumin, 0.01 M sodium succinate, and 0.1 M sodium chloride per mL
Biologic activity	500 MU-I/vial	100 MU/vial	100 MU-M/vial	100 MU-A/vial	1/2.5/10 kMU-E/vial
Molecular weight (kDa)	~400	150	150	900	~700

*Modified From 1) Field M, Splevins A, Picaut P, et al. AbobotulinumtoxinA (Dysport), onabotulinumtoxinA (botox), and incobotulinumtoxinA (xeomin) neurotoxin content and potential implications for duration of response in patients. *Toxins*. 2018;10(12):535; 2) Dressler D, Benecke R. Pharmacology of therapeutic botulinum toxin preparations. *Disabil Rehabil*. 2007;29(23):1761–1768.

The use of escalating doses of botulinum toxins was becoming common practice until safety concerns were raised and fueled by mandates from the FDA. Responding to reports suggestive of systemic spread of botulinum toxins, in 2009 the FDA required new label warnings and a risk mitigation strategy that requires clinicians to discuss the risks and provide written material that details the warnings. The current experience of many clinicians is that using dosages of inco- and onabotulinumtoxinA as high as 600 to 800 units (U) is effective and safe.^{227,264} Two comprehensive reviews concluded that higher doses of BoNT-A appeared to be efficacious in reducing spasticity of the upper and lower limbs after stroke, with minimal adverse effects.^{13,227} In 2017, Wissel et al.²⁶³ reported on the safety and efficacy of escalating doses of incobotulinumtoxinA up to 800 U. The REPAS scale improved significantly. The proportion of subjects achieving at least three of four preidentified treatment goals increased with higher doses of the toxin. No neutralizing antibody was detected.

Dilution. It is believed that increasing the volume of botulinum toxin solution injected magnifies its therapeutic effects by facilitating the toxin's ability to reach more motor end plates. This has been demonstrated in animal studies,^{136,234} where muscle paralysis and atrophy were greater when a more dilute preparation (i.e., higher volume relative to dose, or lower concentration) of botulinum toxin is injected. Humans studies are equivocal in demonstrating superiority of higher volumes of botulinum toxin injections^{81,150} largely because of methodologic limitations of studies, although some investigations have found that high-volume or end plate–targeted botulinum toxin injections result in more profound neuromuscular blockade and spasticity and

cocontraction reduction, as compared with low-volume, non-end plate–targeted injections.¹⁰⁰ As much as high-volume injections appear attractive, it may be a double-edged sword in that it may facilitate distant spread of the toxin. Cases have been reported wherein patients with poststroke spasticity who receive large dilution volumes in proximal upper limb muscles developed transient weakness in the noninjected contralateral upper limb. Based on electrophysiologic abnormalities documented after the injection, weakness was attributed to neuromuscular blockade.

Techniques to enhance toxin effectiveness. There is a lot of interest in techniques to enhance the clinical effects of botulinum toxin without concomitantly increasing the risk of adverse events. Prohibited by recommended total toxin dose and cost, clinicians continuously seek ways to extend the clinical effects of toxin through modification of injection technique and use of adjunctive modalities. Although no standard currently exists, spreading the toxin within a muscle by using a higher volume/more dilute toxin solution (see earlier) and injecting at multiple sites within a muscle are regarded as ways to increase effectiveness by spreading the toxin to reach more muscle end plates, where neuromuscular junctions lie. It has also been suggested that targeting the muscle innervation zone, not just the motor points, can optimize the effects of botulinum toxin administration.¹⁰⁴ In some muscles these areas where intramuscular nerve endings are distributed may also result in better outcomes.¹²¹ Other techniques used to attempt enhancement of toxin effectiveness include guided injection by listening to electromyography (EMG) activity, identifying motor points through electrical stimulation, or visualizing target sites by ultrasonography. The superiority of one guidance technique over

another is yet to be established,¹⁰⁸ but studies have consistently demonstrated that EMG,²⁰⁸ electrical stimulation, or ultrasonography is better than anatomic localization through muscle palpation in improving precision of injections.^{113,201,202,204,233} A potential advantage of ultrasonography is determination of echointensity in assessing changes in muscle architecture related to spasticity, which may affect treatment decision making.¹⁰⁸ Studies are in progress to establish a relationship between echointensity and spasticity.

Adjunctive therapies. Adjunctive therapies are usually viewed as add-on treatment after a more important and impactful modality (e.g., physiotherapy after botulinum toxin injection). As an alternative approach, multimodal therapy²¹⁷ involves the concurrent or sequential use of one or more therapeutic modalities to a current intervention(s) to enhance the overall therapeutic outcome of a concerted spasticity management effort. This is in contradistinction to the concept of adjunctive therapy, which espouses the subsequent addition of one modality to magnify the effect of what is considered the primary treatment.

There is high-level evidence to suggest that adjunct therapies may improve outcomes after botulinum toxin injection.¹⁸⁰ The beneficial role of various adjunctive therapy modalities in enhancing clinical outcomes of botulinum toxin chemodenervation has been reported. Casting can enhance the effect of onabotulinumtoxinA⁶⁹ because prolonged stretching of spastic muscles after botulinum toxin chemodenervation affords long-lasting therapeutic benefit. Extracorporeal shock wave therapy (ESWT) has been shown to have a greater magnitude of enhancement of botulinum toxin enhancement than electrical stimulation, most likely thorough modulation of muscle rheology and neurotransmission.^{117,203,229} The role of adhesive taping is controversial.^{127,226} A commonly used technique to magnify the clinical effect of botulinum toxin chemodenervation is pairing it with superficial electrical stimulation, which influences activity of synaptobrevin-2 receptors that facilitate neuronal binding and subsequent uptake of botulinum toxins.^{15,115,166,258} However, the best stimulation protocol has not been established.²⁰³

Reasons for poor response to botulinum toxin chemodenervation. Botulinum toxin is widely regarded as an effective treatment of spasticity, but the response to treatment varies among persons. In most cases, spasticity relief is not sufficient, while in others even relatively small doses may result in weakness and other adverse events. Table 24.6 summarizes potential reasons for poor outcomes of botulinum toxin chemodenervation.

Immunoresistance. In addition to unintended toxin diffusion to neighboring nerve endings and to distant areas caused by yet

to be defined mechanisms,^{219,247} other undesirable effects of botulinum toxin therapy include sustained blockade of neural transmission that results in prolonged effects similar to anatomic denervation, such as muscle atrophy, and immunoresistance. The latter accounts for loss of responsiveness to the spasmolytic effects of botulinum toxins. Bioassay of neutralizing antibodies (NAB) to botulinum toxin is considered the gold standard in confirming immunoresistance, but more practical clinical tests, such as the frontalis antibody test or unilateral brow injection, can be easily performed by injecting a small dose of botulinum toxin in either the frontalis or corrugator muscle.³² Another method to assess nonresponse is the extensor digitorum brevis (EDB) test that involves measurement of compound muscle action potential amplitude elicited by electrical nerve stimulation of the peroneal nerve before and after injection of a small dose of botulinum toxin into the EDB.¹³²

Based on early reports in the cervical dystonia population, high doses and frequent injections of botulinum toxin were identified as risk factors for immunoresistance.^{101,275} This also provided support for the practice of allowing no less than 90 days in between exposures to botulinum toxins. A much higher incidence of antibody formation has been associated with cervical dystonia than in spasticity.¹⁸⁶ There is growing interest in incobotulinumtoxinA, which is free of excipient proteins and, as such, may have a lower propensity to induce an immunogenic response relative to the other botulinum toxin preparations with complexing proteins.^{17,126} In systematic review that included 1234 participants (91% with stroke) from 14 studies published between 2002 and 2018, NAB prevalence was about 2.6% with no difference among different toxins.¹⁶⁴ However, there are heterogeneous reports. In children with CP, high NAB frequencies of up to 30% have been described.¹¹² The most significant risk factors for antibody formation were frequent treatment and high dose per treatment and a short interval between injections. In a more recent cohort of patients with different neurologic impairments, 83 of 596 patients (13.9%) had measurable NAB.¹ The probability of developing antibodies increased with repeated treatment and was influenced by the BoNT-A formulation. The NAB rates were similar for aboBoNT-A and onaBoNT-A (6% and 7%, respectively), while no NABs were observed in patients treated exclusively with incoBoNT-A. Disease entity and treatment duration had no additional influence.¹ It is thus recommended to avoid booster injections or short-interval injections to reduce the risk of NAB induction.

It is of paramount importance to note that presence of NAB does not equate to nonresponse to treatment.¹⁶⁴ In this review, among the 32 NAB-positive participants, 23 (72%) were nonresponders, but the treatment still seemed effective in 9 (28%). Among the 41 nonresponders, 23 (56%) were NAB positive and 18 (44%) were NAB negative. In other words, the identification of nonresponse by NAB positivity had sensitivity of 56% and specificity of 99.6%. This complex relationship between antibody titers and clinical outcomes of BoNT-A injection is true in other patient populations, such as dystonia.¹¹¹

Repeated injections. In clinical practice, many patients receive multiple injections over a period of many years, sometimes decades, and the long-term effects are not systematically documented. The fact that patients continue to receive botulinum toxin over a long period of time implies that the patients continue to benefit from the treatment without experiencing adverse events. Most studies involving the use of botulinum toxin for spasticity involve only a few cycles of injection. A rare few have reported safety and sustained efficacy up to five injection cycles

TABLE 24.6 Potential Reasons for Poor Outcome of Botulinum Toxin Chemodenervation

Patient Related	Injector Related	Drug Related
- Age and size - Disease condition - Concurrent medications that interact with spasmolytic drugs or alter muscle tone - Unrealistic goals - Immunoresistance - Genetics (?)	- Incorrect diagnosis - Incorrect muscle selection - Improper injection technique	- Incorrect dose - Incorrect preparation - Inactive medication

over a few years.^{66,93,143} Santamato et al.²²⁸ reported safety and efficacy of repeated high-dose incobotulinumtoxinA (up to 840 U/session over eight sets of injections) over a 2-year follow-up period. In a real-world registry study,⁶⁸ high adherence (at least three sessions) to repeated onabotulinumtoxinA injections was observed over a 2-year period in a subset of patients.

Although these few studies claimed that repeated injections were safe, concern remains about the long-term effect of botulinum toxins on muscles. An animal study concluded that the contractile properties of target and nontarget muscles did not fully recover within 6 months of botulinum toxin injections treatment protocol.⁷⁹ The same investigators also found that after repeated botulinum toxin injections, muscle atrophy sets in and contractile material is replaced by fat.⁸⁰ Recognition of botulinum toxin's effects on muscle length and force²⁴⁸ is also emerging, although how this translates clinically is still unclear. These concerning findings need to be investigated further in clinical studies emphasizing muscle changes in recovery after botulinum toxin chemodenervation.

Another area that warrants further investigation is the determination of when other treatment interventions should be considered after repeated botulinum toxin injections. For example, how many botulinum toxin injections need to be done to address a spastic-dystonic “clenched fist” before surgical release of the finger and thumb flexor tendons should be entertained? How many times should a person with spastic paraplegia receive botulinum toxins before intrathecal baclofen therapy is considered? The economic impact of these clinical decisions will also need to be weighed to better appreciate the cost effectiveness of spasticity interventions.

Duration of action. The duration of effect of BoNT-A is widely thought to be approximately 3 months, although a pharmacodynamic model suggests that a single injection may extend up to 6 months.¹⁹¹ Clinically observed duration of BoNT-A effect varies between patients, depending on various factors, such as dose used and clinical condition of the patient.¹⁸ The recently FDA-approved daxibotulinumtoxinA or cervical dystonia appears to have a longer duration of effect⁵⁰; a similar trend was observed in upper limb spasticity in a phase 2 investigation.¹⁹⁶ Clearly, this is one aspect of botulinum toxin therapy that needs further elucidation.

Early treatment. How soon after spasticity onset can botulinum toxins be safely and effectively administered is not well established. A few studies reported that treatment as early as 3 to 6 months of disease onset effectively manages muscle hypertonia and decreases risk of later complications, such as contracture development.^{73,114} A recent study showed that botulinum toxin injection at the emergence of spasticity, as early as 18 days post-stroke, can effectively prevent development of joint contracture. At the same time, muscle weakness related to botulinum toxin did not affect the course of motor recovery.¹⁵⁹ Results of an exploratory, double-blind, randomized placebo-controlled trial²²¹ using abobotulinumtoxinA 500 U in subjects with upper limb spasticity within 2 to 12 weeks poststroke suggested that early treatment significantly delayed time to reach reinjection criteria when compared with placebo. In a retrospective analysis of clinical practice of botulinum toxin treatment, higher doses of toxins were used in the early-start group (on average, 6.4 weeks from stroke onset) compared with the late-start group (49.6 weeks from stroke onset); however, much longer intervals were needed for the subsequent injections (23.1 weeks in the early-start group vs. 14.6 weeks in the late-start group).²⁶⁷

Recovery of function. The effect of botulinum toxin is primarily in decreasing muscle hypertonia caused by spasticity and other

related conditions, such as rigidity. However, there are unusual, but not rare, cases when the outcome of botulinum toxin injection surpasses this expectation and results in increase in functional abilities, particularly when appropriate therapeutic exercise is included in the treatment regimen. Although an obvious explanation for the outcome in this particular case was the uncovering of latent finger and thumb extensor strength once flexor spasticity is controlled, alternative mechanisms should be entertained. Could it be that botulinum toxin merely controlled spasticity of the finger flexors, and it was the ensuing “forced use” and task-specific exercises that accounted for the increased function? Natural recovery may be dismissed because the improvement occurred almost 4 years after the onset of neurologic disease, but it may be postulated that controlling spasticity may have allowed the progression of natural recovery that was stalled when spasticity set in. Did altering muscle hypertonia and stretch properties also modify afferent input into the spinal cord and subsequently influenced motor output?¹⁸¹ Or could it be that botulinum toxins have yet to be defined direct central nervous system effects, as suggested by some reports?^{7,56,90}

Therapeutic weakness. Botulinum toxin injection to spastic muscles concurrently results in both spasticity reduction and muscle weakness, the latter either as a latent phenomenon uncovered by tone reduction or a direct consequence of the toxin. We have observed that, in some patients, when this weakness is addressed by task-specific exercises, it could result in improvement of voluntary control of the weakened muscles. For this reason, we coined it “therapeutic weakness.” This phenomenon of therapeutic weakening is revealed in a recent case of improved voluntary grip control after BoNT-A injection.⁴³

This concept is further supported by another study¹⁴⁹ in which 15 stroke survivors with finger flexor spasticity and residual finger extension received botulinum toxin injections to the finger flexors, and electrical stimulation to the finger extensors after injections for 4 weeks (5 days/week). Finger flexor spasticity was significantly reduced, and finger extensor strength and active hand function were significantly improved as well at 2 and 6 weeks postinjections. It is well documented that botulinum toxin injections effectively reduce spasticity, pain, and abnormal positioning; however, botulinum toxin injections are unlikely to improve active upper extremity function, such as reaching and grasping.^{6,236} The finding that therapeutic weakness of spastic flexor muscles is associated with functional improvement is important in that it can reshape the goals of using this therapy. In particular, therapeutic weakness using botulinum toxin could be considered in stroke survivors who have residual voluntary extension in the upper extremity. Improvement in voluntary control of extensor muscles likely resulted from reduced reciprocal inhibition from the spastic flexors after injection. Previous results have shown that injections can paralyze afferent fibers⁷⁴ and block acetylcholine release presynaptically at neuromuscular junctions.

Focal Pharmacologic Treatment—Nerve Block (Phenol Neurolysis and Cryoneurolysis)

Phenol neurolysis. Before the introduction of botulinum toxins, nerve block using either alcohol or phenol was the only option for focal spasticity management. It was used in the 1950s to chemically ablate nerves to manage cancer-related pain, and it was subsequently applied to relieve spasticity. Over time, percutaneous nerve blocks using phenol or alcohol proved to be effective in controlling focal spasticity across different populations, including CP, TBI, and stroke.^{48,134} Phenol 5% to 7% and

CASE 24.3

Delayed Recovery of Function

PJ is a 69-year-old, right-handed male who presented with a 3-year history of multiple cerebral insults. He was involved in a motor vehicle crash that resulted in a moderate TBI. During the difficult extrication from the vehicle, he lost blood that resulted in hypotension on the scene. While recovering from these in the intensive care unit, he had a brainstem stroke and a suspected hypoxic event. Upon recovery, he had proximal muscle weakness and profound difficulty with walking. He subsequently developed spasticity in the right upper limb, characterized by a clenched fist, involving severe flexion of the fingers and thumb, which prevented his caregivers from cleaning his hand and trimming his fingernails (Fig. 24.5A). He was on oral baclofen 10 mg three times daily at the time of presentation. On examination, he had limb weakness (at best 4/5 in most limbs) and Ashworth score of 4 in the right finger, metacarpal (MCP) and thumb flexors, and 1 in the wrist and elbow flexors. The third distal interphalangeal joint was hyperextended. Given the focality of the spasticity, the clinician suggested weaning the oral baclofen with intent to discontinue after about 4 weeks and botulinum toxin injection of the flexor digitorum superficialis (onabotulinumtoxinA 100 U) and profundus (75 U) and flexor pollicis longus (25 U). At around 1 month after injection, the Ashworth score of the finger and thumb flexors decreased to 2. The patient received another injection of onabotulinumtoxinA 3 months after the first injection to the same muscles using the same doses. An additional 20 U were injected into the lumbricals to address the MCP flexion spasticity. About 1 month later, the patient reported not only a significant decrease in tightness but also a new ability to actively extend his fingers and open his hand (see Fig. 24.5B–D). He also received occupational therapy that focused on forced-use activities and functional retraining. Subsequently, he recovered full active movement of the right hand and the ability to perform functional tasks, such as picking up objects, including utensils and a drinking glass during meals, combing his hair, and buttoning his shirt.



• **Fig. 24.5** Delayed recovery of function. A 69-year-old, right-handed male who presented with a 3-year history of multiple cerebral insults. (A) Spasticity in the right upper limb, characterized by a “clenched fist,” involving severe flexion of the fingers and thumb. (B–D) After weaning from oral baclofen with intent to discontinue after about 4 weeks and botulinum toxin injection of the flexor digitorum superficialis (onabotulinumtoxinA, 100 U) and profundus (75 U) and flexor pollicis longus (25 U), the patient reported not only a significant decrease in tightness but also a new ability to actively extend his fingers and open his hand.

alcohol 35% to 60% denature proteins in neural tissues, leading to blockade of nerve transmission. In addition, phenol appears to result in degeneration of muscle spindles²⁶⁶ and damage to both afferent and efferent nerve fibers.²² This chemical denervation is thought to be irreversible and lead to permanent control of spasticity; however, this is not commonly observed clinically because

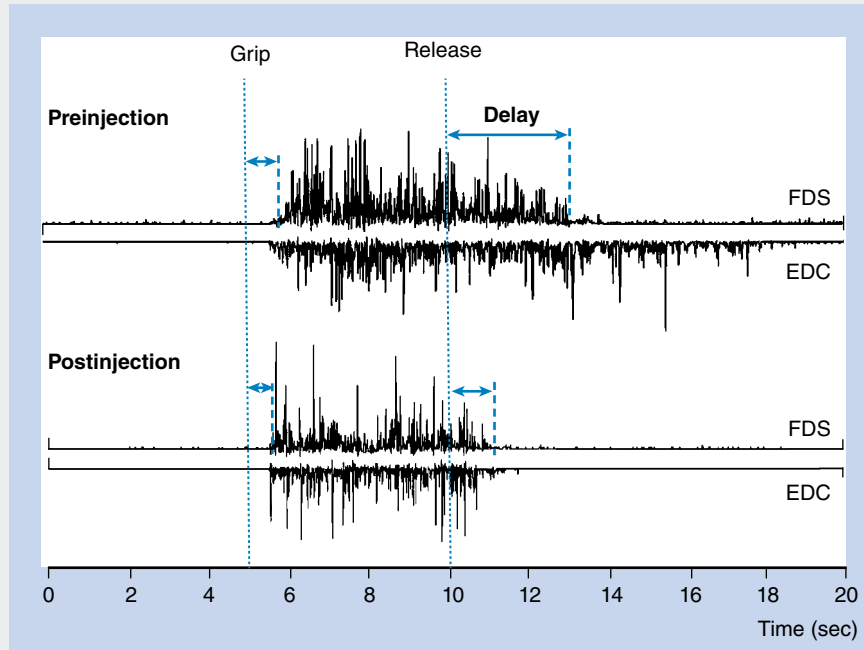
spasticity tends to return several months after the percutaneous block. This may be explained by partial nerve regrowth and sprouting.

Percutaneous injections can either be at the nerve or motor branch level, guided by electrical stimulation or ultrasound. Because phenol is also an anesthetic, especially at concentrations less

CASE 24.4

Therapeutic Weakness

TP is a 53-year-old, right-handed female who sustained a hemorrhagic right middle cerebral artery stroke 3 years earlier. She had finger flexor spasticity and residual weak finger/wrist extension. She received 50 U BoNT-A injection in both the left flexor digitorum superficialis and the flexor digitorum profundus. As expected, botulinum toxin injection led to tone reduction and weakness of the spastic finger flexors; however, she was able to open her hand more quickly probably because of improved ability to more quickly release handgrip. She also reported an increased ability to engage the impaired hand when performing various activities of daily living. This correlated with reduced finger extensor muscle EMG activity (Fig. 24.6). The improvement in voluntary control of hand opening and grip release was likely realized by a decrease in the abnormal cocontraction of finger flexors during voluntary finger extension. This case suggested that a reduction in finger flexor spasticity after botulinum toxin injection is associated with improved voluntary control of residual finger extension.



• **Fig. 24.6** A representative electromyography of flexor digitorum superficialis (FDS) and extensor digitorum communis (EDC) before and 10 days after botulinum toxin injection. The subject was asked to grip as soon and as hard as possible after the “grip” signal and relax after the “release” signal (dashed lines). The release delay time decreases after injection, along with shortened EDC activities. (From Chang SH, Francisco GE, Li S. Botulinum toxin (BT) injection improves voluntary motor control in selected patients with post-stroke spasticity. *Neural Regen Res.* 2012;7[18]:1436-1439.)

than 3%, muscle relaxation commonly observed immediately after injection is caused by the anesthetic effect. The neurolytic effect may not set in until a few hours later because it takes time for the effects on neural tissues to develop. Commonly injected upper limb nerves are pectoral, subscapular, and musculocutaneous. In the lower limb, the obturator and tibial nerves and their branches are commonly treated. However, with advances in guidance techniques using real-time ultrasound imaging and electrical stimulation, phenol neurolysis can be performed to almost all motor nerves or motor branches in upper and lower limbs.^{128,129} Care must be exercised in injecting nerves with significant sensory component to mitigate risk of developing postinjection dysesthesia. Other side effects include localized swelling and excessive weakness. Inadvertent intravascular injection or systemic absorption may result in cardiovascular or central nervous system effects, including hypotension and tremor or convulsions, respectively. In a recent article, we summarized 293 procedures of phenol neurolysis over 3 years in our institution.¹²⁸ Overall, phenol neurolysis has a relatively favorable safety profile, including pain (4%), swelling and inflammation (2.7%), dysesthesia (0.7%), and hypotension (0.7%).

Cryoneurolysis. Cryoneurolysis has decades of established efficacy in the analgesic literature.¹²⁰ Cryoneurolysis applied to motor nerves or motor branches is an innovative intervention for spasticity management. The cryoneurolysis procedure begins with identifying a target nerve under ultrasound guidance. Similar to phenol neurolysis, a cryoprobe is inserted percutaneously through a recommended angi catheter after a weal of lidocaine is placed subcutaneously. The cryoprobe placement is confirmed when electrical stimulation at low intensity (<1 mA) is able to stimulate the target nerve and not the surrounding muscle. Cryoneurolysis is realized through the Joule-Thomson effect¹²⁰ (i.e., throttling a gas through an orifice from a high- to a low-pressure area within the cryoprobe), leading to a drop in temperature at the cryoprobe tip. The temperature is determined by the boiling point of the cryogen, carbon dioxide, or nitrous oxide, typically between -60 and -88°C for pain or spasticity.¹²⁰ At this temperature, an ice ball with a diameter between 3.5 and 18 mm is formed in seconds along the distal cryoprobe by freezing the surrounding fluid. No fluid passes from the probe into the body. The ice ball causes a local zone of axon and myelin disruption leading to a secondary

axonotmesis and Wallerian degeneration of the targeted nerve, extending bidirectionally outward from the lesion. A distinct feature is that cryoneurolysis does not cause damage to the basal lamina, epineurium, and perineurium of the targeted nerve, which serves as conduit for neural regeneration. The axon will regrow after 3 to 6 months, depending on the size of the nerve.¹¹⁶ With cryoneurolysis, there is no dispersion of fluid away from the ice ball toward unwanted targets because the ice ball freezes in place and melts after a treatment cycle is completed. A cycle can last from seconds to minutes; a 106-second cycle uses a handheld Iovera device,²⁵⁹ and a 60-second cycle produces analgesia and hypoesthesia in animals.¹⁸⁵ Blood vessels are typically not affected by the cryoneurolysis at the temperatures utilized. They serve as a heat sink and are not thrombosed.⁸⁷ Winston et al. have demonstrated feasibility and effectiveness of cryoneurolysis in spasticity management through case reports and case series in recent years.^{161,260-262}

Side effects of cryoneurolysis include localized skin effects, paresthesias, and numbness. Most are considered mild and limited to a few weeks to months.^{199,214} Cryoneurolysis causes immediate damage to the nerve. Patients may experience burning and cramping and a painful dysesthesia during the procedure. In clinical experience, cryoneurolysis is more painful than other spasticity treatments, but it ends quickly because the nerve is fully lysed. The first analysis of side effects on 277 nerves in 113 patients found 96.75% of nerves had no side effects after the treatment.²⁵⁹ Among 113 patients, there were three skin events that healed. Nine patients reported nerve pain or dysesthesia (in 3.25% of the 277 nerves; two motor nerves and seven mixed motor sensory targets) with four patients requiring a medical treatment for this. All patients reported resolution on follow-up. Three patients' symptoms remained at 3 months but resolved by 6 months. No patients withdrew from the study because of side effects. There is no maximal dose because no drugs are administered. Further clinical trials on its efficacy are needed.

Nerve blocks using phenol and alcohol have taken a backseat to chemodenervation with botulinum toxins, which appear to have a better safety profile and are easier to administer. Despite its favorable safety and efficacy profile, botulinum toxin treatment alone is often inadequate to fully control symptoms. Typical insurance coverage and FDA approval for spasticity limits the use of onabotulinumtoxinA to 400 to 600 U no more than every 3 months. In those with moderate to severe spasticity, botulinum toxin treatment may not be effective even with a high dose of toxins. Nerve blocks with phenol and alcohol and cryoneurolysis can provide alternative treatment options or can be used in combination with botulinum toxin treatment.

Table 24.7 presents a comparison of the clinical characteristics of botulinum toxins, phenol neurolysis, and cryoneurolysis.

Intrathecal Therapies

As early as the 1950s, intrathecal phenol was used to reduce pain and spasticity. Its popularity has waned owing to its many potential complications, such as sensory, respiratory, bladder, bowel, and sexual dysfunction, and the introduction of baclofen, which appears to have a better adverse effect profile. The use of intrathecal phenol is now limited to specific situations where severe spasticity cannot be managed through any of the other currently available techniques.¹²⁵ Other drugs have been administered intrathecally primarily to manage pain, among which are opioids, gabapentin, and clonidine; the last one has a possible spasmolytic effect.¹⁷⁵ Continuous intrathecal baclofen (ITB) was introduced in the 1980s to treat spasticity in SCI. Since then, several trials demonstrated similar

beneficial effects in other patient populations, such as CP,² brain injuries,¹⁷³ stroke,¹⁷² and MS.²⁰⁰ In addition, ITB was shown to benefit stroke survivors with severe spasticity by decreasing pain and improving quality of life.⁵⁵ A recent series of articles from an expert panel has specifically advised on best practices for ITB therapy, including patient selection,²³¹ screening test,²⁷ dosing and long-term management,²⁶ and troubleshooting.²³⁰

The physiologic effect of ITB involves a decrease in mono- and polysynaptic reflex responses¹⁴⁷ and reduction in resistance to passive stretch.²³² At the cellular level, ITB's mechanism of action is similar to that of the oral form; however, it has the advantage of more direct access to GABA-B receptors in the spinal cord because the blood-brain barrier will not have to be traversed. Thus intrathecal administration allows greater hypertonia reduction and reflex inhibition at doses lower than the oral form, thereby decreasing the risk of adverse events.

ITB is usually considered only when less invasive management options are ineffective in controlling spasticity, although utilization within the first few months of disease onset may be warranted in situations where spasticity of the lower limbs is so severe that waiting for other less invasive options to take effect may predispose to more complications. An investigation on the use of ITB within 3 months of acquired brain injuries showed significant reduction of spasticity and spasm frequency at 1-year follow-up, and their global outcome measures did not differ from those who received ITB therapy later (between 3 and 6 months of disease onset).²¹⁰ Another situation where very early use has been shown to be beneficial is in persons with severe dysautonomia after a brain injury.⁸³ Concern exists that early use of ITB may retard recovery because of the inhibitory effects of GABA,³⁰ but a more recent study suggests otherwise. In an animal model, ITB treatment was initiated at 1 week after closed TBI and maintained for 1 month. It appeared to block the early onset of spasticity and significantly diminish late-onset spasticity and anxiety-like behavior, without causing any significant adverse effect on cognitive and balance performance of rats. Concurrently, it was observed that there was a marked upregulation of GABA, norepinephrine, and brain-derived neurotrophic factor expression in spinal cord tissue.²⁵

The intrathecal system consists of a programmable pump that has an accessible drug reservoir that is implanted subcutaneously in the abdominal area. A catheter that is connected to the implanted pump's side port is tunneled under the skin and enters the spine at the lumbar level. The catheter tip is advanced cephalad and usually left at the thoracic level, although it is not uncommon for it to be placed at the cervical area³ or sometimes intraventricularly.²⁴⁹ Before surgical implantation of the catheter and pump, a trial is performed to confirm ITB's effectiveness in decreasing spasticity, spasms, or pain. The trial typically involves bolus administration of baclofen into the intrathecal space through a spinal needle. An alternative method is to continuously infuse baclofen intrathecally through an external catheter.¹⁰⁷ Given risks associated with this technique, performance of the trial usually requires hospital admission, as compared with the former method, which is carried out in the ambulatory setting. Superiority of one over the other has not been established, although the former appears to be more commonly employed because it is less labor or resource intensive. A successful screening trial is generally defined as an Ashworth scale reduction of 1 to 2 points; however, in certain situations, the Ashworth scale is not always applicable. For instance, the indication for considering ITB may be painful, intermittent spasms, where use of the Ashworth scale as an assessment tool and outcome measure is not valid. Sometimes, the

TABLE 24.7 Comparisons Among BoNT, Phenol Neurolysis, and Cryoneurolysis

	Botulinum Toxin (BoNT)	Phenol Neurolysis	Cryoneurolysis
Mechanism of Action	Blockade of acetylcholine release into neuromuscular junction	Nonselective protein denaturation of nerves and muscles at the injection site	Axonotmesis and Wallerian degeneration from cold temperature without affecting surrounding tissues
Injection Technique	Target muscles, using anatomic landmarks, intramuscular EMG, electrical stimulation or sonographic guidance	Target nerves, preferably motor branches, using electrical stimulation and sonographic guidance	Target nerves, preferably motor branches, using electrical stimulation and sonographic guidance
Onset	3–7 days	Immediate, about 60% of peak effect	Immediate
Peak Effect	2–3 weeks	1–2 weeks	Not yet established
Duration	3 months	Variable, >3 months in general	Variable, >3 months in general
Dose Titration to Desired Effect	Yes	Yes	Usually 1–3 cycles per nerve or nerve branch
Ease of Administration	Relatively easy	Requires advanced training and expertise	Requires advanced training and experience
Evidence of Efficacy	Robust, RCTs published	Several RCTs	No RCT, ongoing trials
Pain During Injection	Some	More	More
Pain Days After Injection	Rare	More	Rare, 96.75% of 277 nerves had none in a recent study
Adverse Events	Less common	Higher incidence of dysesthetic pain and swelling (<4%)	Negligible for motor nerves; whole mixed motor sensory nerves have increased incidence
Use During Pregnancy	Not recommended. Category C drug per FDA	Not recommended. Birth defects observed in animals, but not reported in humans.	Not contraindicated
Cost	Expensive	Very affordable	Significantly less than BoNT

EMG, Electromyogram; FDA, US Food and Drug Administration; RCT, randomized controlled trial.
Adapted from Li S, Winston P, Mas MF. Spasticity treatment beyond botulinum toxins. *Phys Med Rehabil Clin N Am.* 2024;35(2):399-418.

indication is to decrease tone (regardless of Ashworth scores) just enough to facilitate positioning for care and transfers or to allow participation in a therapeutic exercise program. In addition to the usual assessment utilizing the Ashworth scale and ROM testing, videotaping while performing tasks actively or passively may help clinicians, patients, and caregivers determine whether intrathecal infusion of baclofen may help with spasticity or spasms. The recommended intrathecal bolus dose of baclofen during screening trial is 50 µg. If ineffective, the option is to try higher doses sequentially: 75 and 100 µg. Some patients, such as those with MS and incomplete SCI, present with profound weakness confounding spasticity and may experience exaggeration of the underlying weakness immediately after the ITB screening trial. First, the patient must be assured that weakness during the trial is reversible and that it does not mean that the same degree of weakness will occur after ITB pump implantation. The finding of increased postscreening trial weakness merely demonstrates that ITB has the potential to abolish spasticity and suggests that the patient will most likely require a low dose of ITB after pump implantation. Thus instead of starting with the “usual” starting dose after pump implantation (determined by doubling the dose used during the screening trial that resulted in significant decrease in spasticity), the starting daily dose may be the same bolus dose used during the trial. To illustrate, if a patient experienced good

control of spasticity or spasms with a 50-µg bolus of baclofen but also experienced an increase in weakness, the initial dose after pump implantation may be 50 µg infused continuously over a 24-hour period. Another strategy to assuage concerns about excessive weakness during the screening trial is to use a lower dose (e.g., 25-µg bolus). Although 97% of patients with spasticity resulting from SCI¹⁹⁸ and 94% with cerebral origin spasticity⁹¹ respond positively to bolus screening trials, it is still advantageous to perform the screening trial to allow the patients and caregivers an opportunity to experience a state of low or no spasticity. The trial results can also guide the physician regarding future dose targets and a patient’s peculiar response to intrathecal baclofen.

When planning ITB therapy, nonmedical conditions need to be considered, such as the patient’s ability to comply with dose titration and pump management, which will require regular visits with the clinician. Management does not end with implantation of the pump; frequent dose adjustments will be needed subsequently to reach appropriate ITB dose. We define an optimal dose as one that adequately controls (not completely abolishes) spasticity without causing any unwanted effect. Thus optimal dose can be determined only by titrating the dose upward until a side effect occurs (e.g., excessive weakness, bladder retention, constipation) and then decreasing to a level that does not cause the adverse event. The frequency of regular refills of the pump reservoir

depends on the dose and concentration of the drug. The importance of patient and caregiver education regarding the need for pump maintenance, including refills, and recognition of potential ITB system-related problems cannot be overemphasized.

CASE 24.5

Intrathecal Baclofen Withdrawal

LY is a 27-year-old female in a minimally conscious state secondary to anoxic brain damage 10 years earlier, who presented to the emergency room with generalized seizures. Before this, her seizures have been well controlled with carbamazepine. Her mother reported that a few days earlier the patient “just did not seem like herself” and that since the day before seemed “a lot tighter in the legs than usual.” The mother reported that the patient’s urine seemed to be cloudier, but did not pay much attention to this. The mother had difficulty pointing out the exact day she noticed these changes in the patient because of her daughter’s inability to meaningfully communicate. Past medical history is significant for seizure disorder that developed after the anoxic event but has been well controlled by carbamazepine. An ITB pump was implanted 7 years earlier to better control spasticity and extensor posturing. Current ITB dose is 974.8 μg /day, which controls lower limb spasticity well enough to facilitate care and transfers, and the most recent intrathecal pump refill was 6 weeks earlier. Aside from carbamazepine and baclofen, the patient was not on any other medication. At the emergency room, she also had a temperature of 102.1°F. Blood pressure was 120/70 mm Hg, and heart rate was 110 beats/min. The mother added that the patient seemed more restless and uncomfortable. The patient was subsequently admitted to the critical care unit, where temperature increased to 104°F. Efforts to decrease hyperthermia using a cooling blanket and acetaminophen failed to resolve fever. While awaiting the blood and urine test results, including culture, she was empirically started on broad-spectrum intravenous antibiotics because the clinician was suspecting sepsis that may have triggered the seizures. Chest radiograph was normal, and a noncontrast brain computerized tomography did not show any acute lesion. Six hours after admission to the unit, she had status epilepticus despite intravenous lorazepam and phenytoin. Temperature climbed to 106.8°F. Urinalysis, complete blood count, and electrolytes were normal. Later that evening, the patient went into cardiac arrest.

The accompanying case demonstrates the importance of recognizing the signs and symptoms of ITB withdrawal, especially when they appear concurrently with other abnormalities. A suspicion of sepsis-triggered seizure in this case was justified because of the report of changes in urine character and fever. However, there were other earlier signs that should have tipped about ITB withdrawal, such as recurrence of spasticity (“...a lot tighter in the legs than usual...”). This, plus the nonspecific change in the overall condition (“...just did not seem like herself...”), sudden recurrence of seizures, and fever, should have raised the suspicion of ITB withdrawal in its advanced stage. The ITB pump should have been interrogated to assess function, a plain radiograph of the pump and catheter should have been done to assess gross catheter integrity, and intrathecal or enteral bolus of baclofen should have been considered. ITB withdrawal evolves over 1 to 3 days⁴⁹ and usually results from an abrupt decrease in GABA activity. Supportive measures and drugs, such as benzodiazepines, intravenous dantrolene,¹³⁵ and cyproheptadine,¹⁷⁴ may help mitigate withdrawal symptoms, but the definitive treatment is restitution of ITB delivery. Fig. 24.7 presents an algorithm for troubleshooting suspected ITB problems.

The most common drug-related side effect is hypotonia, followed by somnolence, headache, convulsion, dizziness, and urinary

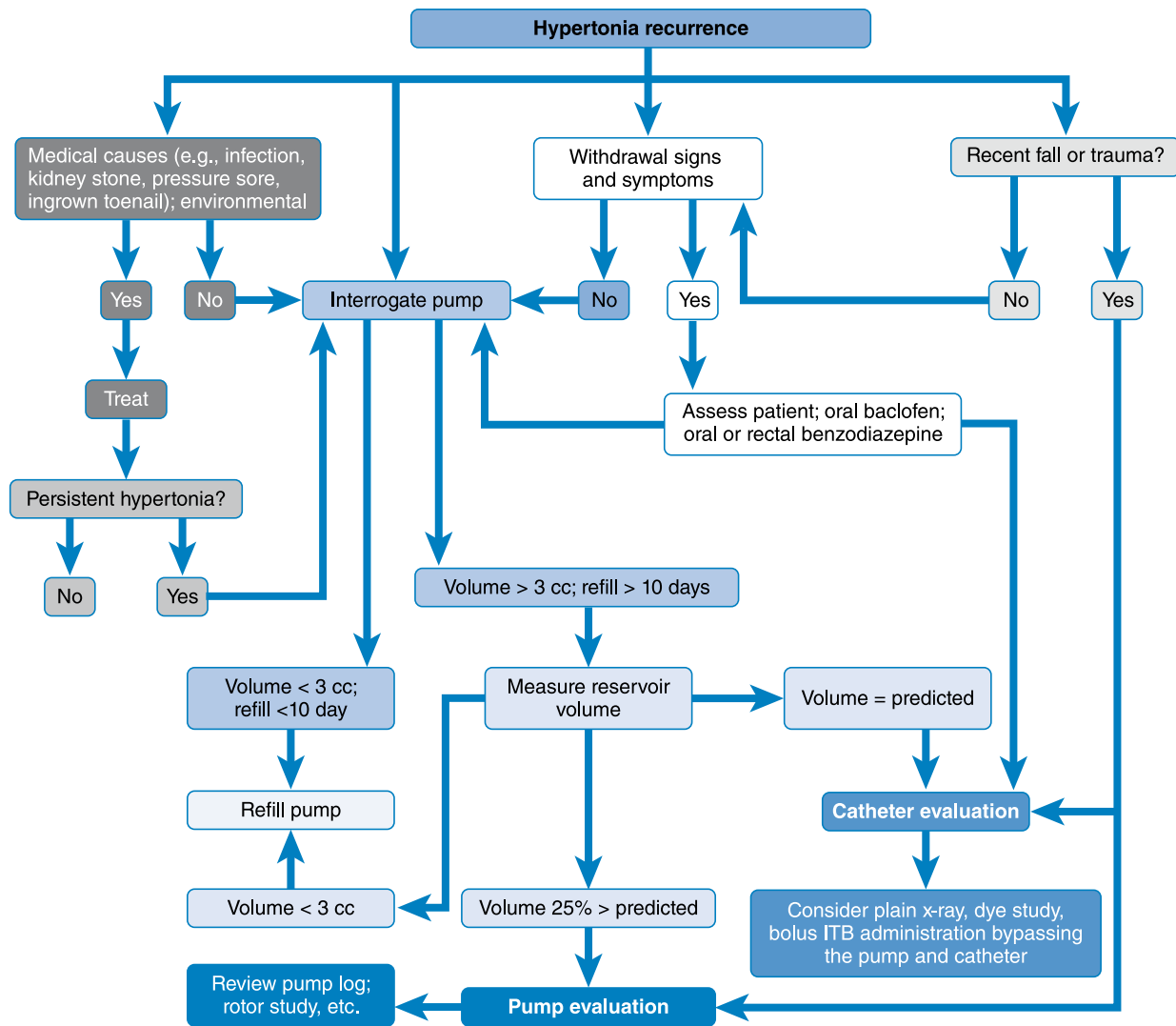
retention. Pump-related adverse events include pump stall, but catheter-related problems, including dislodgement, fracture, and kinking, are more common. Finally, clinician-related complications, such as dosing and programming error that can lead to either under- or overdose, can occur.

Surgical Intervention

Surgical management of spasticity is a well-accepted treatment option for contractures because it primarily addresses joint deformities rather than spasticity itself. Surgical interventions include neuroablative procedures, such as peripheral neurotomies and dorsal rhizotomies, and orthopedic reconstructive procedures, such as tendon lengthening and tendon transfer. Surgical intervention is a permanent treatment. Optimal management of spasticity by nonsurgical means should be attempted before surgical treatment is employed. Therefore it is often viewed as a treatment of last resort. However, surgical intervention could be pursued earlier in cases where the underlying problem is best treated with surgery or when other options are unavailable because of lack of availability or resources.⁸⁹ When excessive spasticity and contracture are not sufficiently controlled by therapy and pharmacologic treatment, tendon lengthening is often considered. This involves correction of abnormal joint posture alignment, allowing for improved ability to move joints and in many cases facilitate activity and exercise.¹³¹ Common tendon procedures include split anterior tibial tendon transfer and Achilles tendon lengthening to manage spastic equinovarus.⁶¹ Tendon lengthening and release can also help with upper limb management.⁵ Tendon lengthening could be performed with minimal invasive technique via percutaneous needle tenotomy.⁵³ Surgical interventions primarily manage joint deformities rather than spasticity itself. Tendon lengthening elongates the tendon and subsequently corrects the abnormal joint position, but it does not change the contractile and mechanical properties of muscle and its innervation. In other words, the unchanged spastic muscle has a new resting position at a corrected joint position via the elongated tendon. The reflex responses are both velocity and muscle (not tendon) length dependent¹⁵⁵ and related to muscle fiber length and sarcomere numbers.^{77,85,157} The exaggerated reflex responses are still expected to remain unchanged if the same stretch is applied with reference to the new resting position. Therefore tendon lengthening can improve posture but is not likely to correct altered stretch reflex in spastic muscles. Other procedures, such as neurectomy, hyperselective neurectomy, neurotomy, and hyperselective neurotomy, have also been shown to benefit spastic conditions.^{8,37,148,176}

Emerging Therapy—Hyaluronidase Injections

Advances in understanding pathogenesis of poststroke spasticity (see Fig. 24.1) not only allow a better understanding of the relation between spasticity and motor recovery but lead to the development of new treatments as well. One example is the use of human recombinant hyaluronidase to address the active muscle stiffness component of spastic hypertonia.²¹⁶ The accumulation of hyaluronan within spastic muscles is considered the primary extracellular mechanism promoting development of muscle stiffness. It was then hypothesized that hyaluronidase, the enzyme of hyaluronan, is expected to reduce muscle stiffness if it is injected into the muscles.²¹⁶ In their pioneering study, Raghavan et al. injected human recombinant hyaluronidase with saline in 20 patients with unilateral upper limb spasticity at a single visit. The



• **Fig. 24.7** Algorithm for assessing suspected intrathecal baclofen (ITB) withdrawal syndrome. (Redrawn from Francisco GE, Saulino M. Intrathecal baclofen management of spasticity. In: Brashear A, ed. *Spasticity: Diagnosis and Management*. 2nd ed. Demos Medical Publishing; 2016:287-298.)

outcome measures included safety, passive and active movement, and muscle stiffness at eight upper limb joints. These were assessed at four time points: preinjection (T0), within 2 weeks (T1), within 4 to 6 weeks (T2), and within 3 to 5 months postinjection (T3). No clinically significant adverse effects were observed from the injections. Passive movement at all joints and active movement at most joints increased at T1 and persisted at T2 and T3 for most joints. The modified Ashworth scores also declined significantly over time postinjection. This study provides preliminary evidence that hyaluronidase injections are safe and potentially efficacious for muscle stiffness reduction. This treatment is very promising in that it reduces muscle stiffness without producing weakness and increases voluntary movement of the treated limb. However, further research is warranted, such as a large-scale clinical study, dosing, duration of effects, and the effect on neural component.

Other Emerging Therapies

In addition to the various treatment options described, other modalities are emerging as potential primary or adjunctive spasticity management options, many of which need to be further

evaluated systematically. These include acupuncture²⁶⁹ and electroacupuncture,⁴⁰ vibration,²²³ dry needling,^{170,224} and noninvasive brain,^{14,103,141,268} spinal,²⁰⁵ and transcutaneous¹⁷⁹ nerve stimulation. The effect of peripheral nerve stimulation appears inconsistent.^{123,140} ESWT is also being investigated.^{192,270} Currently, there is very little research evidence or clinical experience to support the use of these interventions, and as such they are not widely used in clinical practice. Elsewhere, provocative preclinical studies have been published recently on interesting aspects of spasticity management, including the role of calcium intake on spasticity¹²² and potential prevention of spasticity development after experimentally induced spinal injury and nerve transfers with or without neurectomy.^{124,255} High-frequency electrical motor nerve block¹³⁸ and deep brain stimulation¹⁵⁸ are also being investigated as potential treatments.

Summary

Spasticity is a commonly seen with many neurologic conditions. While the term “spasticity” chiefly refers to muscle tightness in response to stretch, it has become a catch-all word to describe the

clinical phenomenon of the upper motor neuron syndrome, including other impairments that lead to hypertonia and dysfunction. Recent developments suggest that the clinical observations of “spasticity” are not solely from central nervous pathology but also may be a consequence of concomitant and consequential intramuscular changes. Data acquisition and physical examination should go beyond spasticity assessment at rest, but take into consideration the consequences of spasticity on well-being and function. This, in turn, will inform the formulation of treatment goals, which should be personalized and mutually agreed upon by the clinician and patient. There are multiple pharmacologic and non-pharmacologic treatment options. A multi-modal management approach is recommended for best clinical outcome. The resurgence of the popularity of surgical treatment may reflect the recently-revived recognition of expanding treatment options beyond the algorithmic stepwise approach. Emerging treatments for spasticity include new medications (mucosal, injectable), cryoneurolysis, various physical modalities and neuromodulation interventions.

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